CHAPTER

Gender in agriculture and food systems

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1 Introduction

Both women and men are involved in agriculture throughout the world, although the roles they play differ significantly by region and are changing rapidly. Gender shapes access to productive resources and opportunities, cutting across many assets, inputs, and services—land, livestock, labor, technology, education, extension, and financial services, with women having less access to these across a wide range of contexts (Quisumbing et al., 2014). These gender differences in resources and opportunities shape the agricultural sector, both within smallholder agricultural farming systems and within larger commercial systems. Thus, to understand agriculture, we must understand the gender dynamics within agriculture.

Understanding gender in agriculture, however, begins by understanding the difference between "sex" and "gender," terms that may be confusing to researchers and practitioners because they are often used inconsistently and interchangeably. Sex refers to the innate biological categories of male and female and is a fixed category

rooted in biological differences. On the other hand, gender refers to the social roles and identities associated with what it means to be a man or a woman (or neither) in a given society or context. Gender roles may be shaped by ideological, religious, ethnic, economic, and cultural factors and are a key determinant of the distribution of responsibilities and resources between men and women Moser, 1989; gender distinctions are not necessarily binary. Because gender roles are socially, rather than biologically, determined, they are fluid and subject to change based on changing norms, resources, policies, and contexts. "Gender" and "women" are often—but inaccurately—used interchangeably. Gender refers to relations between men and women, not an exclusive focus on women. Because much of the literature and practice of agricultural development has focused on men, an attention to gender often means rebalancing and paying more attention to women but should not be to the neglect of men. Gender also intersects with other domains of disadvantage, such as poverty, caste, ethnicity, age or life-cycle stage, and the like. Thus, while it is often used as an analytical category, it should not be taken in isolation from other factors that affect development outcomes.

Analyzing gender in agriculture also means going beyond the sex of the household head. The sex of the household head is a faulty gender indicator because it ignores the majority of women in the world who live in households defined as male headed as well as men who live in households headed by women (Doss, 2018). Thus, it confounds gender issues with those of household structure; a woman is only considered the head of the household if there is no adult man present or no adult man who is economically contributing to the household. Some analyses now distinguish dual-headed households from those that are headed solely by a man or a woman. It may also be inappropriate in multigenerational households to define one person as the head; different adults may have different roles and responsibilities. At the micro level, gender analysis involves looking at a broader range of individual characteristics such as age and status within the household, as well as looking at women within male-headed households.

Because gender relations are complex and vary across contexts, nuanced and context-specific data collection and analytical methods are recommended, often involving both qualitative and quantitative methods in the same study (Behrman, Meinzen-Dick, & Quisumbing, 2014). In this chapter, owing to the primarily quantitative audience of the Handbook, we focus on quantitative data collection and analysis, but also highlight where qualitative methods have enhanced the quantitative studies. We focus on studies that employ gender analysis, not studies that focus on women only. Gender analysis examines how the social roles of men and women are determined and how these roles affect the outcomes being studied. In agriculture, gender analysis provides insights into how socially constructed roles and responsibilities shape the many decisions from agricultural production, processing, participation in markets, all the way to consumption and well-being outcomes.

This chapter critically reviews and assesses the rapidly growing literature on gender and agriculture, focusing on empirical studies from low- and middle-income

countries. The first part of the chapter deals with models and measurement. Section 2 begins by reviewing household models of production and consumption, contrasting models that assume efficient household allocations with those that do not. We discuss the implications of complex household structures, the neglect of jointness of household decisions, and incomplete risk-sharing within the household. Section 3 then discusses implications of advances in household models for measurement and data collection, focusing on recent developments in measuring assets, decision-making, empowerment, and time use.

The following sections apply gender analysis to production, markets, and consumption and well-being outcomes. Section 4 reviews gender and agricultural production, starting with gender gaps in agricultural resources, agricultural productivity, and the gender dynamics of technology adoption. Section 5 goes beyond the household to examine gendered participation in markets, including modern supply chains. It also presents emerging evidence from evaluations of interventions to improve gender equity in marketing schemes. Section 6 examines how women's empowerment and gender equality affect nutrition outcomes, and how gender dynamics affects the takeup and impact of nutrition-sensitive agricultural programs (NSAP). Section 7 concludes and explores areas for future work.

2 Understanding rural household behavior

Research on gender in agriculture has benefited from the growth in the theoretical and empirical literature on household models. Since the 1980s, the dominant model of household behavior has been the unitary model, which conceptualizes households as groups of individuals who have the same preferences and fully pool their resources (Becker, 1981). Drawing on the theoretical work by Chiappori (1988, 1992) and Bourguignon, Browning, Chiappori, and Lechene (1993) empirical work in the 1990s has shifted this concept of the household in which households decide "as one" to a "collective" model in which individual household members may have different preferences, may not completely pool resources, and may bargain over outcomes in both production and consumption (Alderman et al., 1995). Key to the collective model is the assumption that allocations within the household are Pareto-efficient. More recent work on household behavior has expanded tests of the collective model and questioned the efficiency assumption. Because tests of household models have informed our understanding of gender in rural households, particularly in developing countries, we briefly review the literature, focusing on empirical studies related to gender and agriculture.[1]

[1]There is a large empirical literature on intrahousehold allocation that does not necessarily focus on outcomes related to gender and agriculture. Helpful reviews can be found in Strauss and Thomas (1995), Behrman (1997), and Doss (2013).

2.1 Modeling efficient vs. inefficient households

2.1.1 Unitary models

The unitary model characterizes the household as a group of individuals who behave as though they were a single individual. Either all members of the household share the same preferences or a (self-interested or altruistic) dictator makes all the decisions. Consider a household consisting of two individuals, a man (m) and a woman (f), both of whom derive utility from a consumption bundle of commodities including goods and leisure ($\mathbf{x}$), conditional on household characteristics ($\boldsymbol{\gamma}$).[2] The household utility function can be expressed as:

$$U(\mathbf{x}; \boldsymbol{\gamma}) \tag{1}$$

which is maximized subject to an income or budget constraint:

$$\mathbf{p} \cdot \mathbf{x} = Y = y_J + y_m + y_f \tag{2}$$

Total household income comprises joint income (y_J) as well as individual male and female incomes (y_m and y_f, respectively). Since preferences are identical or there is a dictator, utility maximization under standard assumptions leads to a series of demand functions for $\mathbf{x}$ that are functions of prices ($\mathbf{p}$), total household income (Y), and household characteristics ($\boldsymbol{\gamma}$):

$$x_i = x_i(\mathbf{p}, Y; \boldsymbol{\gamma}) \tag{3}$$

The key feature of Eq. (3) for this discussion is that only total household income (Y) is relevant for demand and not its components, i.e., the specific male or female contributions. This result is known as income pooling.[3]

The unitary model has been very powerful in predicting and explaining a variety of intrahousehold phenomena (Becker, 1981). Nevertheless, the assumption of a single household welfare function reflecting the preferences of all members is not innocuous. If individuals do not have the same preferences and there is no dictator, then the different preferences must be aggregated.

2.1.2 Collective models

Concern about the strong assumptions underlying the unitary model has spawned several alternatives that require weaker assumptions. One class of these comprises collective models (Chiappori, 1988, 1992) that allow different preferences for individual household members, assuming only that decisions are made so that outcomes are Pareto-efficient.

Building on the two-person household described above, assume that preferences are altruistic such that the husband cares about the wife's allocation so that her

[2]We denote vectors in bold type (e.g., $\mathbf{x}$ and $\boldsymbol{\gamma}$) and scalars, including elements of vectors, in ordinary type (e.g., x_i).

[3]Note that income pooling in this sense does not require that household members physically combine their financial resources into a single pot or bank account.

private consumption increases his welfare, and vice versa. This leads to individual-specific utility functions in which each member's consumption appears: $U_i(\mathbf{x}_i, \mathbf{x}_j; \boldsymbol{\gamma})$ with i = f, m and j = m, f. Then, for all Pareto-efficient outcomes, there exists some weight μ for which the household's optimization problem can be written as:

$$\text{Max } \mu U_m(\mathbf{x}_m, \mathbf{x}_f; \boldsymbol{\gamma}) + (1-\mu) U_f(\mathbf{x}_m, \mathbf{x}_f; \boldsymbol{\gamma}) \tag{4}$$

subject to the earlier budget constraint in Eq. (2). Eq. (4) shows that the unitary model is a special case of the more general model; the former obtains either when the individual utility functions U_m and U_f are identical (common preferences) or when μ is equal to zero or one (dictator). This formulation yields demand equations that in addition to prices, incomes, and household characteristics as in Eq. (3), are now also functions of the weight, μ:

$$x_i = x_i(\mathbf{p}, Y, \mu; \boldsymbol{\gamma}). \tag{5}$$

The above derivation can be interpreted as a two-stage budgeting process. In the first stage, household members pool all their income and allocate it according to the weight, or sharing rule, μ. This sharing rule is likely to be related to individuals' relative bargaining power within the household; a more powerful individual would command a greater share of the household's resources. In the second stage, each individual maximizes his or her utility given his or her income share.

Letting a_m and a_f represent proxy measures for bargaining power (for the man and woman, respectively) that influence μ, demand functions ignoring price variation can be expressed as

$$x_i = x_i(Y; \mu(a_m, a_f), \boldsymbol{\gamma}). \tag{6}$$

Holding household income constant, the effect of individual bargaining power on demand for good i can be interpreted as the effect of changing the share (μ) of household income allocated to each household member. Because income pooling implies that the identity of the income earner, or person in control of the resources, is irrelevant, these effects should be zero:

$$\delta x_i / \delta a_j = 0 \text{ with } j = m, f \tag{7}$$

This provides a straightforward test of the unitary model by including proxy measures for male and female bargaining power in the estimation of demand equations.

2.1.3 Extensions of the collective model

Later formulations of the collective model have expressed demand functions explicitly as conditional on distribution factors z; a distribution factor is a variable that does not enter individual preferences nor the overall household budget constraint but influences the decision process (Bourguignon, Browning, & Chiappori, 2009). Distribution factors are important because: (1) their existence is inconsistent with the traditional, unitary framework; (2) the influence of distribution factors provides the only testable restrictions for the collective model without price variation; and

(3) they can help recover some features of the intra-household decision process (Bourguignon et al., 2009). Such z-conditional demand functions express the demand for one type of good as a function of the demand for another good, total expenditure, and distribution factors. Empirical tests of the collective model using z-conditional demand functions only require a distinction between factors that affect the allocation process within households (distribution factors) and those that are likely to affect personal preferences. Interestingly, Bourguignon et al. state that the tests do not require any knowledge of the actual intra-household allocation of goods nor any assumption on the nature of the goods that are consumed or produced within the household.

Finding plausible distribution factors remains an empirical challenge. The early literature that tested household models used proxy measures of bargaining power, given the absence of direct measures. These included: (1) shares of income earned by women (Hoddinott & Haddad, 1995); (2) unearned income (Schultz, 1990; Thomas, 1990); (3) current assets (Doss, 2005); (4) inherited assets (Quisumbing, 1994); (5) assets at marriage (Quisumbing & Maluccio, 2003); and (6) the public provision of resources to specific household members (Lundberg, Pollak, & Wales, 1997). These early papers focused on identifying bargaining power in consumption decisions and were largely based on observational data. Later work testing the collective model has drawn from lab-in-the-field experiments, natural experiments, and randomized controlled trials (see Doss & Quisumbing, 2019 for a review).

More recent empirical work using distribution factors uses exogenous variations in resources provided to specific household members. Attanasio and Lechene (2014) use the random allocation of PROGRESA, a conditional cash transfer program in Mexico, and the relative importance of the husband's and wife's networks of relatives in terms of size or wealth as distribution factors. Having data on two plausible distribution factors enables the authors to reject the unitary but not the collective model. Chiappori and Naidoo (2020) state that any variable that affects an individual's position on the "marriage market" is a potential distribution factor, provided that its variations can influence the intrahousehold allocation of resources. This highlights the importance of considering changes in marriage and inheritance laws and legislation affecting property rights and the labor market as potential distribution factors. Early tests of the collective model had limited ability to evaluate the intertemporal effects of policies because they ignored the dynamics of intrahousehold processes. Chiappori, Giménez, Molina Chueca, Theloudis, and Velilla (2020) test three alternative models—full intertemporal commitment, no intertemporal commitment, and limited intertemporal commitment—ranging from one extreme in which spouses fully commit to all future (and state-contingent) allocations of resources between them to the other, in which spouses renegotiate Pareto weights in every period. Using data from the US Panel Survey of Income Dynamics (1999–2017) they find that couples' labor supply is consistent with limited commitment (significant impact of older shocks, cross-effects from past shocks to partner's hours), which differs from the impact of shocks on singles' labor supply. The authors conclude that

"the extent to which individuals commit to their partner for life is crucial for understanding how policy may affect household formation, divorce, income, wealth, and child development among other things" (Chiappori et al., 2020, p. 34), which reiterates the role of marriage markets and institutions, distribution factors that affect outside options.

2.2 Efficiency of household decision-making

2.2.1 Pareto efficiency

The collective model assumes Pareto-efficient allocations. Under the unitary model, the share of income or assets controlled by either partner, given total income, should have no effect on allocations. Under the Pareto-efficient collective model, on the other hand, the sharing rule is affected by each individual's bargaining power.

Using data from France and Canada, Bourguignon et al. (1993) and Bourguignon, Browning, Chiappori, and Lechene (1994) find that the ratio of individual income effects are not equal to one, rejecting the unitary model, but also find that the ratios are constant across a range of commodities, which is consistent with Pareto-efficiency. Similar results have been reported for Taiwan (Thomas & Chen, 1994), Indonesia (Thomas, Contreras, & Frankenberg, 2002), Bangladesh, Ethiopia, Indonesia, and South Africa (Quisumbing & Maluccio, 2003), and Mexico (Attanasio & Lechene, 2014; Bobonis, 2009). These tests of the unitary vs. the collective model have generally found that consumption outcomes are Pareto efficient, but much of this literature focused on dual-adult rural households, not complex rural households.

2.2.2 Inefficiencies in production

Although shifting from the unitary to the collective model of the household has expanded our understanding of household behavior, two areas remain unresolved. First, there are clearly documented cases in which households do not reach efficient outcomes. In these situations, noncooperative bargaining models may be more appropriate: they do not assume that resources are pooled and explicitly allow for outcomes where potential gains from cooperation have not been realized. Individuals may act strategically; each individual makes separate but interrelated production and consumption decisions based on his or her own preferences and interests and expectations of what others will do.

For example, a now classic study in Burkina Faso, where men and women control and cultivate separate plots, finds that total household crop yields could have been increased by shifting fertilizer from men's fields to women's fields (Udry, 1996). Using data from a 4-year panel study of 150 households in 6 villages in 3 agroclimatic zones, Udry estimates yield as a function of plot characteristics (including plot area, slope, and soil type), the gender of the plot manager, a household-year-crop fixed effect, and an error term. The key empirical finding of the paper is that plots controlled by women have significantly lower yields than other plots within the

household planted to the same crop in the same year but controlled by men. This suggests that the household does not behave Pareto-efficiently with respect to production: the household overall could achieve higher output by reallocating variable factors (such as fertilizer) from plots controlled by men to plots controlled by women or by reallocating land from women to men.

Udry's (1996) study spawned a flurry of research on efficiency in agricultural households. Since the original study was based on data from a limited geographical area, a logical extension was testing whether the finding of Pareto inefficiency was robust. Akresh (2009), using nationally representative data, found that only households in regions geographically proximate to those studied by Udry exhibited Pareto inefficient intrahousehold allocations, while the rest of the country revealed no evidence of Pareto inefficiencies. Akresh suggests households in regions experiencing negative rainfall shocks are less likely to have inefficient intrahousehold allocations, possibly because the cost of inefficiency is greater.

Other studies find other instances of productive inefficiency. McPeak and Doss (2006) find that male household heads among East African pastoralists located the household farther from town to limit women's milk marketing.

Subsequent studies on Burkina Faso delve further into decision-making in complex households as sources of productive inefficiency. Akresh, Chen, and Moore (2011) using the same ICRISAT data used by Udry, explore incentives for cooperation among household members in polygynous households. They find that wives in polygynous households obtain higher yields, possibly owing to their being more likely to coordinate agricultural production with their co-wives than with their husbands. The difference in plot yields between household heads and wives is smaller in polygynous households; they attribute this to more efficient production among co-wives rather than between spouses. The authors hypothesize that altruism, in the form of shared public goods (between husband and wife), inhibits cooperation, because altruism places lower bounds on the utility attained in a noncooperative equilibrium. In contrast, individuals such as co-wives, who are not linked by altruistic ties, have greater incentives to cooperate.

Akresh, Chen, and Moore (2016) revisit the issue of greater inefficiency in monogamous compared to polygynous households and develop a model with a husband and two co-wives with differing degrees of altruism toward each other. They show that stronger altruism can encourage players to choose a noncooperative strategy by increasing the utility obtained in the noncooperative equilibrium, therefore reducing both the gains to cooperation and the threat of punishment. The authors point out that these results hold only under the assumption of separate spheres, which they justify given the gender division of labor within Burkinabe households, particularly with respect to childcare. While the authors find that women cultivators have lower yields in general, women cultivators in polygynous households have higher yields, controlling for the number of children. They hypothesize that women in polygynous households may be better able to specialize, possibly sharing both farm and childcare duties, but women in monogamous households cannot do so (Akresh et al., 2016).

Also in Burkina Faso, Haider, Smale, and Theriault (2018) analyze fertilizer and labor use across plots that are collectively and individually managed. While they find evidence of input sharing, they do not find an efficient allocation of fertilizer.

2.2.3 Inefficiencies in consumption

Inefficiency in consumption has been documented in a wider range of countries, partly because it is a less data-intensive test compared to productivity analysis, which requires detailed plot-level data on inputs and outputs, and because a range of consumption outcomes can be analyzed. We review some illustrative studies. Akresh et al. (2016) analyze a 2-year pilot program that randomly distributed cash transfers that were either conditional or unconditional and were given to either mothers or fathers. Countering the conditional wisdom that transfers to mothers lead to better outcomes for child human capital, this study finds that giving cash to mothers does not lead to significantly better child health or education outcomes, and that money given to fathers improves young children's health, particularly during years of poor rainfall. If the household operated efficiently, the identity of the transfer recipient should not matter.

In Bangladesh, Anderson and Eswaran (2009) investigate the relative contributions of earned vs. unearned income in enhancing women's autonomy. Their theoretical model demonstrates that earned income could be more important than unearned income in empowering women; this is confirmed by data from Matlab, Bangladesh. Their study also reveals the finding that employment outside their husbands' farms is what contributes to women's autonomy. Although not an explicit test of Pareto efficiency, this points to the importance of different threat points affecting intrahousehold allocation. While models of household behavior commonly use divorce as the threat point that couples face, their paper provides evidence that noncooperative behavior within marriage is the relevant threat point. This scenario generates predictions consistent with their empirical results; predictions with divorce as the threat option are refuted by the data.

Castilla and Walker (2013) use data from a field experiment in Southern Ghana to test the effect of windfalls on intrahousehold allocation. The results from the experiment indicate that spouses behave according to a noncooperative marital allocation contract with differences in spending patterns by gender and by the degree to which these windfalls are public. Public windfalls received by husbands lead to increased investment in assets and social capital, while there is no such effect when wives win.

2.2.4 Inefficiencies in both production and consumption

There is also evidence of Pareto-inefficiency in both production and consumption decisions within complex households. Kazianga and Wahhaj (2017) explore differences between nuclear and extended families in rural Burkina Faso using nationally representative data from 2010, 2011, and 2012. This study distinguishes between plots that are collectively- and individually-managed. The authors find that nuclear family households are close to being efficient in household production and do not

reject the hypothesis that they engage in efficient intrahousehold risk-sharing in consumption choices. However, extended family households are found to be inefficient in both production and consumption choices.

Also using a nationally representative data set from Burkina Faso, Rangel and Thomas (2019) investigate the conditions under which complex agricultural households are inefficient with respect to both consumption and production decisions. The authors explicitly account for the complexity of household structures, including multigenerational and polygynous households. Using data on budget shares for a wide range of commodities, they find that in farm households headed by a monogamous couple (with no coresident adult sons), resource allocations are consistent with efficiency. In more complex household structures, including polygynous households, efficiency in allocations is not rejected in models that allow more than two household members to have agency in decision-making. In contrast, tests for efficiency based on whether the same farm households maximize profits by equating marginal products across plots are rejected for all household types. Further, these same tests indicate individuals do not equate marginal products across their own plots.

Rangel and Thomas (2019) argue that models of resource allocation based on production decisions may be misleading, owing to measurement error and unobserved heterogeneity at the plot level (Goldstein & Udry, 2008; Gollin & Udry, 2021), or because of market failures that differentially affect plots owned by a household (Jones, Kondylis, Loeser, & Magruder, 2019). Analysis of consumption decisions, particularly in households with many decision-makers, provides better insights into decision-making within complex households.

It is noteworthy that papers using data from Burkina Faso have dominated this literature. Even if there is variation in social norms within the country, one should not expect that these results would hold across all developing countries. Indeed, Lafave and Thomas (2017) use data from multiple waves of the Indonesia Family Life Survey to extend the collective model to examine allocation within the extended family. They analyze the relationship between height for age, cognition, kindergarten attendance of children and the distribution of wealth among extended family members. They find that extended families share resources: child human capital outcomes are affected by wealth of non-coresident family members. While they reject the special case in which all members are completely altruistic, they fail to reject the restrictions of the efficient model, indicating that non-coresident family members coordinate allocation decisions to achieve Pareto efficiency.

Why is studying household models important to understanding gender in agriculture? First, it challenges assumptions that decision-makers within the household have the same preferences. Second, it shows that whoever has more power within the household, which varies across cultures and contexts, may be able to affect the intrahousehold allocation of resources, which has implications for both production and consumption outcomes. Third, it helps to identify the areas in which efficiency failures may have costs, as well as the sources of these failures.

2.3 Unpacking household decision-making

The above literature has shown cases in which the number of decision-makers increases the complexity of decision-making in polygynous households. However, even among households with two decision-makers, the collective approach, which has been strongly influenced by bargaining models, often misses key elements of household dynamics. Bargaining models emphasize competition and rivalry among household members. Yet, even if household members do not completely pool resources, the fact that people form households, share ownership and control over some resources, work together on family farms, produce some output jointly, have and raise children together, and share in some consumption indicates that there are gains to jointness in gender and family dynamics (Doss & Meinzen-Dick, 2015; Fafchamps & Quisumbing, 2007). Understanding both individual roles within households and the levels of cooperation, including joint decision-making and ownership of resources, is essential to understanding agricultural households, especially those that produce and consume their own food. Jointness and cooperation—and possible gains to efficiency and welfare from increasing cooperation—has until recently either been assumed or has been neglected in the analysis of household behavior.

Lab-in-the-field experiments, particularly public goods games, have provided an additional perspective on household cooperation. In public goods games, each player is given an amount of money which they can keep or contribute to the "public" or household pot. The money that is put into the household pot is then increased (often multiplied by 1.5) and then divided among the players. The highest payoff to the household overall occurs if both players put all of their money into the household pot. If, however, one player does not expect the other to contribute to the pot, then their best individual response is not to contribute.

Several studies find inefficiencies in the contributions of husbands and wives. In the simple game where the household pot is evenly split, spouses generally contribute some but not all of their allocation to the household pot, potentially leaving some income on the table (Barr, Dekker, Janssens, Kebede, & Kramer, 2019; Fombe, Sama-Lang, Fonjong, & Mbah-Fongkimeh, 2013; Iversen, Jackson, Kebede, & Munro, 2011; Munro, Kebede, & Tarazona-Gomez, 2014). This does not imply that households never fully cooperate in this sense; instead only that, on average, households do not fully cooperate.

Variations on the game provide further insights. While Iversen et al. (2011) reject the notion that spouses maximize the surplus, they find significant differences across villages in Uganda, suggesting that the patterns are heterogeneous. They alter the allocation rule: in addition to the standard approach where the surplus is split evenly, they introduce an option whereby the control over how to split the surplus is given either to the husband or the wife. They find that when women control the allocation, both husbands and wives contribute more to the household pot. However, there is no evidence that women generally contribute more than men.

Kebede, Tarazona, Munro, and Verschoor (2013) introduce variations where the allocation decision may be given to the husband or the wife. The researchers also ask

spouses what they expect the other to contribute. While husbands' expectations of their wives' contributions are higher than their wives' actual contributions, wives' expectations of their husbands' contributions are lower than what the husbands do contribute. While overall, men contribute slightly more than women, on average, people contribute only about half of their allocation to the common pot, leaving substantial surplus unrealized.

The question remains as to why household members do not take full advantage of the potential gains. Castilla (2015) uses a lab-in-the-field approach to demonstrate that spouses are hiding money from each other. Hoel (2015) use a dictator game to identify different household types. They find that while many households do cooperate and obtain the full benefits, others do not. To identify whether the patterns seen in the lab in the field games are correlated with actual agricultural outcomes, Lecoutere and Jassogne (2019) use a public goods game in combination with an elicitation of consumption preferences regarding the household pot. They find that households that are both more cooperative and are egalitarian in their consumption (neither partner claiming more than their share) are more likely to invest in sustainable intensification of food and cash crops. Analyzing pastoralist households in Northern Senegal, Hoel, Hidrobo, Bernard, and Ashour (2017) demonstrate that households that obtain less cooperative outcomes in the public goods games are the ones largely driving the gender gap in milk production. In these households, cows owned by women produce 10.6% less milk than those owned by men, even after controlling for household, owner, and cow characteristics.

The findings from these experiments suggest that there may be gains to increasing cooperation within the household. Yet, the extent to which the experimental games reflect reality and which interventions would increase cooperation remains unclear. In some cases, what appears to be cooperation may in fact reflect coercion, where the person with less power reports cooperation in order to mask disagreements or because they are afraid to express individual agency.

Interviewing multiple people within the same household also yields insights into decision-making. In Section 3.1, we discuss in greater detail studies that interview multiple household members, who may respond differently to the same question. Most of these studies use observational data where both husbands and wives are interviewed, but others use a more complex survey design where questions on decision-making or on control and ownership of assets are randomly assigned to different respondents within the same household. The results of these studies show varying degrees of concordance and disagreement, which can be quite revealing about intrahousehold dynamics.

Several studies have specifically analyzed questions about decisions and resources relevant to agriculture. Using Tanzanian survey data, Anderson, Reynolds, and Kay (2017) finds that husbands and wives provide different responses regarding whether women are involved in household and farming decisions, and that these vary across the types of decisions. Similarly Twyman, Useche, and Deere (2015) find that, in Ecuador, the reports of women's involvement in agricultural decision-making are lower when the responses are provided by men rather than women.

We are interested not only in whether spouses provide different responses, but also how their responses are associated with outcomes—either in terms of agricultural decisions or welfare outcomes—for the household. Analyzing the relationship between responses and autonomy in Ghana and Bangladesh, Seymour and Peterman (2018) conclude that the level of agreement on decision-making is indicative of the underlying power dynamics within households. Ambler, Doss, Kieran, and Passarelli (2021), also using data on Bangladesh, argue that the different responses from husbands and wives on asset ownership and household decision-making are consistent with asymmetric information within the household. Using both qualitative and quantitative data, Acosta et al. (2020), find that men and women have different understandings of what it means to make a joint decision. Bernard, Doss, Hidrobo, Hoel, and Kieran (2020) explore why a particular individual or individuals are the ones within the household to make a decision.

2.4 Risk pooling and management

Risk is a perennial feature of agriculture (see Chapter "Risk management in agricultural production" by Tack and Yu). In this chapter we focus on the gendered implications of risk pooling and management. Studies of risk-pooling indicate that men and women may manage risk differently, pooling risks in some contexts but not in others. The emphasis of these studies differs: some examine men's and women's responses to shocks (and indirectly test whether risk is pooled within the household), while others investigate whether men and women have different risk preferences, using different methods of eliciting risk preferences.

2.4.1 Risk-pooling

Early studies on risk-pooling, analyzing how household consumption and budget shares were affected by shocks to different sources of income, find that household members do not perfectly share risk (Doss, 2001 on Ghana; Duflo & Udry, 2004 on Côte d'Ivoire). More recent studies extend this analysis by focusing on individual level consumption outcomes, using different measures of shocks, and analyzing impacts on individually- and jointly-owned assets. Similar to the earlier studies, these studies consider the role of social norms in household and individual risk-smoothing strategies.

Villa, Barrett, and Just (2011) examine how individual dietary diversity responds to income shocks among two groups of East African pastoralists. Using detailed panel data on each individual, and distinguishing between positive and negative income shocks, Villa et al. (2011) show that consumption behavior of different individuals responds asymmetrically to shocks. Dietary diversity of household heads exhibits statistically significant responses to below-mean income and appears unresponsive to above-mean income, while wives' and adult sons' dietary diversity is unresponsive to below-mean income but has statistically significant responses to above mean income. Adult daughters' dietary diversity remains relatively stable when household income fluctuates. The authors attribute sons' systematically worse

dietary diversity relative to other household members to sons' primary occupation in herding away from towns, which limits their access to diverse diets. Daughters' slightly better dietary diversity may arise from their higher value owing to the practice of bride price. In this specific context, the authors question the assumption that females are universally worse off, which could lead to males being excluded from targeted food and nutrition programs.

Asfaw and Maggio (2018) examine how social norms regarding land inheritance in Malawi interact with joint or sole management of land in mediating the impact of weather shocks on household consumption outcomes. They find that temperature shocks adversely affect household consumption, food consumption, and caloric intake, but rainfall shocks do not have consistent impacts. When women solely manage household plots, temperature shocks not only affect consumption outcomes, but also impinge on nonfood consumption. These effects are not found in households where men solely or jointly cultivate land, suggesting that households where women are the sole managers of land are more vulnerable. They also find that in matrilineal districts, where women traditionally have more secure property rights, households where women are the sole managers of land are less vulnerable to shocks compared to similar households living in patrilineal districts.

Assumptions about the extent of income pooling also affect the framing of testable hypotheses. Also using data from Malawi, Josephson (2017) tests the assumption that all household income is pooled, accounting for income earned individually by men and women as well as joint income. While she rejects the hypothesis of complete income pooling and full insurance within the household, she finds evidence that household members partially insure one another for expenditure on essential goods (including food, clothing, education, and healthcare) but not for luxury goods (such as cigarettes and alcohol, recreation, and housing and utilities). This result holds only when joint income is included; when only individual incomes are considered, her findings show that household members do not pool income or insure against shocks. This suggests that omitting joint relationships fails to account for an important dynamic in household analysis.

More recent studies take advantage of the availability of comparable sex-disaggregated data on land and assets for different countries to explore how social norms influence households' risk management strategies. For example, Quisumbing, Kumar, and Behrman (2018) examine the effects of covariate and idiosyncratic shocks on solely- and jointly-owned assets owned by men and women in Bangladesh and Uganda. Jointly-held land and assets in Bangladesh were better insured against food price increases in 2007–2008 compared to jointly-held assets and wives' assets in Uganda. The authors posit that differences in the institution of marriage and cultural concepts of joint and individual ownership affect the extent to which joint or individually owned assets are used to cope with shocks. In Bangladesh, joint land and asset holdings do not change significantly in response to shocks while individual assets are sacrificed at the margins, indicating that husbands and wives try not to sell assets that are the economic base of the household unit. In contrast, in Uganda, husbands' assets appear better insured than wives' or even joint assets.

Recent analyses on risks and shocks exploit the increased availability of geo-referenced data on weather-related variables, such as temperature and rainfall, on a sufficiently disaggregated scale that allows mapping to household surveys. The growth in the availability of "big data" will provide opportunities to examine how conditions at the macro- and mesolevels affect outcomes within the household, if the appropriate identifiers are available to link data sources. Together, these studies suggest that we should be analyzing the conditions under which household members pool their risk. The research presented here suggests that social norms and the extent to which household members pool income and other resources will affect how they handle shocks.

2.4.2 Risk preferences

The abovementioned studies document incomplete risk pooling but do not attempt to measure men's and women's risk preferences. Lab-in-the-field experiments are useful tools for eliciting men's and women's risk preferences directly in a variety of settings. For example, Magnan, Love, Mishili, and Sheremenko (2020) use experimentally derived prospect theory (PT)-based risk preferences from couples in Tanzania to test whether men's and women's risk preferences influence the adoption of improved maize varieties. They find that women's risk aversion and men's loss aversion are negatively correlated with improved variety use. The tendency of men to overweight small probabilities is negatively associated with adoption, whereas the tendency of women to do so is positively associated. These results indicate that relying only on the household head's preferences to understand adoption is misguided; technology promoters should consider targeting both men and women in their education and dissemination efforts.

Kramer and Lambrecht (2019) use a framed field experiment with 1527 men and women in Ghana to elicit preferences for investments in crop farming vs. other business activities. The experiment was framed as a rural development program aiming to improve income-generating capacity in targeted areas. Participants were informed that one man and one woman in their community would, through a lottery, be selected to receive an in-kind investment grant. Prior to this lottery, all participants were asked to choose how to invest this grant, and these choices were binding for participants selected to receive the grant. Men and women were interviewed separately to analyze gender gaps in revealed investment preferences. The authors find that, despite strong beliefs that men and women are more skilled in crop farming and nonfarm businesses, respectively, many respondents invest in both farm and nonfarm activities. The small gender gap in revealed preferences for nonfarm diversification is largely explained by gender stereotyping around perceived skills and increasing access to physical and human capital (varied randomly within subjects) does not significantly affect preferences. The authors conclude that both men and women reveal a strong preference for diversified investments, which needs to be reflected in programs and policies aiming to improve women's economic empowerment.

Bageant and Barrett (2017) use 3 years of household survey data, administrative records and qualitative interviews to examine the relationship between gender and

demand for index-based livestock insurance (IBLI) among pastoralists in southern Ethiopia. The authors explore whether risk aversion, informal insurance, product education and female-held assets are relevant to women's demand for IBLI but do not find a robust gender difference in overall demand for IBLI, nor gender differentiated-demand along the four hypothesized dimensions.

Neither do Clarke, De Nicola, Vargas Hill, Kumar, and Mehta (2015) find significant differences between men and women in their demand for agricultural insurance products in an experimental demand-elicitation exercise in rural Bangladesh. However, men were slightly more likely to purchase dry-days insurance during the rain-fed agricultural season than women, but women were overwhelmingly more likely to buy life insurance for their spouses than were men. These highlight not only men's higher engagement in agriculture and their better understanding of main agricultural risks but also women's vulnerability to the death of their spouses, since men are the main providers in their households. In a related study using framed field experiments involving lotteries and a specific insurance contract in rural Bangladesh, Clarke and Kumar (2016) find that, given the opportunity, women are just as likely as men to purchase agricultural insurance. They also find that financial literacy and an understanding of the insurance product affect takeup of these products, disadvantaging women, who have fewer years of schooling and lower financial literacy than men. Without individual data on preferences, characteristics affecting takeup, and the context (framing) of relevant decisions, we could erroneously conclude that women are "always" more risk-averse than men.

These studies on risk preferences illustrate the importance of eliciting preferences directly from men and women and challenging the assumption that women are always more risk-averse than men. These cases show that gender differences in risk aversion may depend on the sphere of decision-making, the relative vulnerabilities of men and women, and gender gaps in physical and human capital.

3 Data and measurement

To analyze the agricultural sector and to understand how rural transformation is affecting both women and men, gender disaggregated data are needed. While there is other information in this volume on best practices on data collection (see Chapter "Agricultural data collection to minimize measurement error and maximize coverage" by Carletto et al.) this chapter focuses on the data needed for gender analysis.

First, microlevel data needs to be gender-disaggregated; data must be collected at the level of the individual, rather than just at the household or farm level. Second, data are needed for researchers to analyze how institutions and structures, such as markets for inputs and outputs, credit markets, and labor markets, are experienced differently by men and women. This may involve data collection on institutions, markets, and value chains in addition to individual and household level survey data.

Broadly, we can consider four types of surveys relevant to agriculture: farmer surveys, multipurpose household surveys, labor market surveys, and agricultural censuses. Each provides opportunities and challenges regarding collecting gender-disaggregated data.

Farmer surveys typically interview individual farmers, often focusing on those who grow a particular crop. The farmer is usually defined as the person who makes the major agricultural decisions or the person who knows the most about the agricultural production. The survey asks detailed questions about the production, processing, and sales and marketing of the particular crop. Often, the details on the one crop means that survey questions give scant attention to the other farming and non-farm activities of the farmer. Although women farmers have become more visible, in many places women on smallholder farms do not necessarily see themselves—and are not seen by others—as farmers. Instead, they are seen as farm helpers or farm wives. Lists of farmers drawn from producer groups or those who receive extension services may be biased toward men farmers who are more visible and more likely to participate in these institutions. Thus, explicit efforts are needed to ensure that women farmers are included.

Multipurpose household surveys with detailed agricultural modules, such as the Living Standards Measurement Study (LSMS), are often used to analyze smallholder farming systems.[4] These approaches acknowledge that agriculture is embedded in a set of broader household activities and decisions and collect data on both production and consumption. They provide a better means for understanding livelihood decisions but may have less detailed information on specific crop production practices. These surveys often rely on proxy respondents (see Chapter "Agricultural data collection to minimize measurement error and maximize coverage" by Carletto et al.), often the male household head. Thus they often do not interview women, who may manage their own plots, crops or livestock or make a substantial contribution to agricultural production and may influence agricultural decisions. It is important to count both men's and women contributions and to identify the opportunities and constraints that they face.

The FAO recommends that countries conduct an agricultural census every 10 years. This involves complete enumeration of all agricultural holdings in a country. While the coverage is comprehensive in terms of agricultural holdings, the amount of data on each holding is fairly limited. The holder is identified as the person, typically the male household head, who manages the agricultural holdings (for smallholder farms, this is the household farm). If data is also included on the subholdings, which are parts of the holding managed by a particular person in the household, then there is greater scope for gender-disaggregated data.

[4]The LSMS (https://www.worldbank.org/en/programs/lsms) has collaborated with national governments to collect household survey data from a wide range of low and middle income countries. Two recent sets of surveys provide more gender disaggregated data. The LSMS-Integrated Surveys on Agriculture collect detailed gender disaggregated data at the plot level. The LSMS+ surveys interview multiple people within the household so that both men's and women's voices are represented.

Finally, labor force surveys collect data on the economically active population, employment, and unemployment. Because they ask about employment in the agricultural sector, they provide aggregate statistics on the agricultural workforce. Data are typically collected on all household members of working age. These data are used to report the share of the population working in agriculture and to examine trends in the agriculture workforce. It is important to ensure that questions are asked to identify whether women are engaged in agriculture; in many instances, they may be reported as a homemaker, rather than a farmer, even when they undertake agricultural production.

3.1 Who should be interviewed?

Collecting gender-disaggregated data involves collecting data about individuals within the household. It is not sufficient to collect data on the farm enterprise because we also need to know who is involved in the different components. In many large-sample surveys, a proxy respondent, typically either the household head or the person considered most knowledgeable, provides information about everyone in the household. A proxy respondent may be asked the "who" questions about many issues: who within the household owns the land, makes the decisions, or provides the labor. If the responses to these "who" questions can be linked to the household roster, then the analysis can go beyond gender to consider other characteristics such as age, education level or marital status (Doss & Kieran, 2014).

Yet, there are many reasons why a proxy respondent may not be appropriate. For topics such as preferences over crop characteristics, aspirations, or knowledge regarding agricultural practices or legal rights to resources, we are seeking the respondent's personal views and the individual concerned must be interviewed.

In addition, interviewing more than one person in the household may be necessary to obtain data on the full range of agricultural production undertaken by the household. One person, even if he or she is the head of the household, may not know about the full range of activities being done or the full range of constraints that household members face. The head of household cannot provide this information for other household members.

When multiple people within the household are asked the same question, they often provide different responses to questions about asset ownership (Ambler et al., 2021; Kilic & Moylan, 2016), the value of assets (Doss et al., 2018), labor force statistics (Bardasi, Beegle, Dillon, & Serneels, 2011), financial services (Fletschner & Mesbah, 2011), and household decision-making, both regarding agricultural decisions (Anderson et al., 2017) and broader household decisions (Donald, Koolwal, Annan, Falb, & Goldstein, 2020). Having conflicting responses makes the analysis more difficult. At times, it may be useful to reconcile the responses. For example, data on the marital property rules in a country combined with information on how and when property was acquired may enable the researchers to determine whose response matches the legal definitions of ownership. In other situations, it may be useful to understand their different perspectives (Ambler et al., 2021).

While most of these analyses about different responses have used observational data collected from multiple household members, several use an experimental design to identify how the choice of respondent affects the responses. To analyze responses regarding household asset ownership in Uganda, Kilic and Moylan (2016) assign households to one of five treatment arms. They find differences in reported levels of asset ownership depending on who is interviewed and the structure of the interview. Similarly, a study in Ecuador that uses an experimental design of interviewing with three treatment arms where the respondent is (1) the husband, (2) the wife, and (3) both the husband and wife, obtains very different responses regarding decisions on pest and farm management (Alwang, Larochelle, & Barrera, 2017).

A final issue is whether the data is being collected to analyze gender implications, comparing the outcomes for men and women, or for intrahousehold analyses. For the former, it may be sufficient to have information on men and women from different households. This may simply mean ensuring that both men and women farmers are interviewed. For the latter, it is necessary to have information on men and women within the same household. If only the spouses in the principle couple are interviewed, the contributions of other adult household members may not be counted. The right approach will depend on the research question being asked and the local context.

3.2 Scope of questions

In this section, we focus on the specific data needed for gender analysis within agricultural studies. As noted above, the key to collecting gender-disaggregated data in surveys is to identify "who" controls resources, provides labor, and makes decisions, with best practice to interview multiple people within the household.

Best practice begins with a household roster listing all household members and some of their relevant characteristics. Then the ID code for the household member(s) can be listed in the appropriate sections. The "who" can then be linked not only to the gender of the person but also other characteristics, such as age or marital status. This approach allows for a deeper understanding of intersectionality and how gender interacts with these other individual characteristics.

Basic demographic data are needed on all the people involved in agricultural production, including the holder and subholders of the land, those who provide paid and unpaid labor on the holding, and those involved in making decisions. The minimum demographic data needed are sex, age, education level, marital status, and relationship to household head or respondent for each of these people.

All these characteristics may affect an individual's involvement in agricultural production and their access to the necessary resources. Marital status may affect status within the community and access to land and social networks. Dual-headed households may have more access to resources than those headed by sole women. Widows and women who are divorced may have less access to a variety of resources, especially land. Even when women are defined as the farmer, their role as decision-makers may vary, depending on whether they are married.

3.2.1 Asset ownership

Much of the data on asset ownership continues to be collected at the household level, rather than at the individual level, although there is increasing evidence on both the importance and feasibility of disaggregating this data by gender (Doss, Deere, Oduro, & Swaminathan, 2015). Detailed discussions of how to collect asset ownership for gender analysis are available (Doss, Kieran, & Kilic, 2020; UNDESA, 2019).

Data on asset ownership may be collected using either individual men and women or the asset as the unit of analysis. For the former, each person in the relevant population is identified as an owner or a nonowner. Indicators can be created for the proportion of men or women who are owners and for the proportion of owners who are women. The second approach is to identify the owner of each asset. This is appropriate for relatively large assets, such as land, housing, businesses, large livestock, consumer durables and large agricultural equipment. The owners may be identified as men, women, or joint.[5] We can then consider the distribution of the assets by the form of ownership. This indicator is particularly relevant to understand the gender distribution of assets.

Ownership may not be the only relevant factor. It may also be useful to know whether a woman, for example, has the control or use of assets. The woman in the household may have a mobile phone that she can use, even if she does not own it herself. This might mean that she can call in for information, but that she is not the one who receives calls or texts.

For some types of assets, such as livestock, different individuals may hold the various rights. For example, one person may have the right to obtain and sell or consume the milk, but may not have the right to sell or slaughter the animal (Njuki, Kaaria, Chamunorwa, & Chiuri, 2011). Thus, depending on the purpose of the data collection, it may be important to understand who holds these different rights.

Finally, data on how assets were acquired may also be useful. Rights associated with land may differ depending on the mode of acquisition. Purchased land may have a more complete set of rights, including the rights to sell and mortgage, than land that was inherited or allocated under customary tenure. In places with a partial community property marital regime—one in which property acquired during marriage is considered to be jointly owned, but property brought to the marriage is not—understanding when and how the property was acquired will affect who has the legal rights. There may be differences among men and women regarding how assets are acquired. (See Quisumbing, Payongayong, Aidoo, & Otsuka, 2001 for Ghana and Doss, Truong, Nabanoga, & Namaalwa, 2012 for Uganda.)

[5]Various categories of joint may be considered, depending on the context; joint could include spouses, parents and children, siblings, or those inside and outside of the household.

3.2.2 Land rights

Land is typically the most complex of the assets on which we want to collect data. The best approach is to collect information on each parcel of land. The owners of each parcel should be identified, along with information on whether their name is on any form of documentation.

To collect information on the use and control of land, two types of questions are used. The first is questions about the rights over the land, including the right to sell, bequeath, rent out, improve, or collateralize the land. For gender analysis, it is useful to both know whether someone in the household holds the right, and if so, who holds it. Women may not be able to use the land as collateral because there are no sources of credit based on land for either men or women. This situation is different from one in which only men can use the land as collateral.

A second type of question asks about decision-making with regard to the land. This allows for distinctions between who owns, manages, and controls the output. The decision-making questions often include information on who decides what to grow, what inputs to use, and how to manage the output. Many recent surveys, including many of the LSMS Surveys, identify each of the relevant decision-makers. Slavchevska, Doss, de la O Campos, and Brunelli (2020) find that for six African countries, these three categories do not necessarily overlap; the owner is not necessarily the manager or the one who controls the output.

Tenure security may influence agricultural production decisions. In some contexts, tenure security may be associated with land ownership or long-term leases or rental markets. In other places, tenure security may be within a customary tenure system. An extensive literature exists on conceptualizing and measuring tenure security.

Three components of tenure security were identified by Place, Roth, and Hazell (1994): the extent to which the various rights are held by one person or persons; the duration of the rights; and the robustness of the rights, particularly whether they are recognized and enforceable. Doss and Meinzen-Dick (2020) add a fourth dimension which is the extent to which the rights are held individually or shared with others.

Joint ownership does not necessarily imply equal claims. Recent studies in rural Uganda and South Africa found that many women were reported as jointly owning land with their husband. Very few of those households, however, had any form of ownership documents. In addition, very few of the women's names were on the documents and women claimed fewer rights over the land than men (Doss, Meinzen-Dick, & Bomuhangi, 2013; Jacobs & Kes, 2014).

While there is ample evidence that tenure security matters for agricultural productivity at the household level, little has explicitly considered it at the individual level. One of the sources of tenure insecurity for women is the death of their husband or divorce. Peterman (2012) documented that across 15 African countries, widows inherit no property, often resulting in a trajectory into poverty. Dillon and Voena (2018) find that married couples in Zambian villages where widow's inherit invest more in their land than couples in areas where women do not inherit, suggesting that secure tenure for the women influences household decisions. The relationships

between tenure security and investments in land are complex. In Benin, the demarcation of lands as part of a titling program resulted in increased fallowing of demarcated (and thus more secure) land, which increases soil fertility, particularly for women headed households who tend to be more tenure insecure. Yet, it also resulted in a transfer of some investments, particularly those which are used to claim land rights, to land that was not demarcated (Goldstein, Houngbedji, Kondylis, O'Sullivan, & Selod, 2018).

3.2.3 Labor

Labor is the second important input into agricultural production. New approaches allow better collection of labor input into agriculture (see Chapter "Agricultural data collection to minimize measurement error and maximize coverage" by Carletto et al.).

Yet the undercounting of women's labor is not simply an issue of measurement. Deere (2005) identifies four reasons why women's labor may be undercounted in population and agricultural censuses, based on research in Latin America. First, even when actively engaged in agricultural production, rural women are likely to report their home as their principal occupation. Second, censuses ask about income-generating activities, thus undercounting subsistence production. Third, the definition of agriculture is often narrow, focusing on fieldwork, leaving out a range of predominantly women's activities, including raising livestock, tending kitchen gardens, and agricultural processing. Finally, when definitions of economic activity involve a minimum number of hours of work in the week prior to the survey for certain categories, such as for unpaid family workers but not for the primary farmer, women are often undercounted due to the seasonality of agricultural work.

At a minimum, if agricultural surveys ask about the various tasks being done, they should disaggregate by age and sex. Men's and women's labor has often not been interchangeable in agricultural production, but the patterns of activity are changing and data are needed to document this. Women are increasingly involved in many traditionally male tasks, especially as men migrate out of rural areas. Men are typically involved in activities that have a higher return, regardless of whether the crops were traditionally women's crops.

In addition to details on employment and agricultural work, collecting information on time use reveals on how people allocate their time across productive and reproductive activities and leisure. This provides a means to analyze the trade-offs across these various sets of activities. As the agricultural sector transforms, time-use data are critical to understanding how people allocate their time within and outside the agricultural sector. Measuring time use is discussed in more detail below.

3.2.4 Income

It is relatively easy to collect individual level income data when people are involved in wage work. Collecting income data when people are self-employed is more challenging, but the attribution of income is still clear for a one person-enterprise.

It is more challenging to assign income to individuals when they work in household enterprises, whether agricultural or nonagricultural.

Full income, including both cash income and the value of the goods and services produced and consumed within the household, is a useful measure of well-being. Income measures are typically used to categorize households—and sometimes individuals—as poor or nonpoor. Income from agricultural production is often not collected in a way that allows the attribution of any of the income to individuals. Surveys often ask who controls the income from agricultural production. Attributing household farm income to individuals is not simply a measurement problem, but is also a conceptual problem. When men, women, and children all contribute labor to the farm, how should we assign the income across the individuals?

If we are interested in intrahousehold dynamics, the various sources of income, or women's status and income-generating opportunities, then collecting information on all sources of income by individual may be important. Without attention to this detail, we may easily ignore women's many small sources of income. These smaller sums of money may have important implications for the well-being of the women themselves, since their control of this income may determine how it is spent. Thus, policies that affect these sources of income might have a relatively large impact on household well-being or expenditures on particular types of goods, but these relationships cannot be identified if data is collected only on the major sources of income.

3.2.5 Institutions

Finally, to understand the different opportunities and constraints men and women face, it is useful to have data on their participation in a range of agricultural institutions, especially those delivering services. Women typically have had less access to extension services and agricultural credit (see Ragasa, 2014). Extension is moving increasingly from the government sector to the private sector, including civil society organizations, and it is important to know how this change is affecting women. Although microcredit has often targeted women, larger-scale agricultural credit is still often out of their reach. Data on which individuals access these resources are needed to understand who is being left out and to track programs to provide such resources. This would include whether they have contact with extension workers, participate in farmer's groups, obtain credit, and sell produce in the market. Simply knowing who participates is not sufficient to prescribe how to ease the constraints, but it provides useful baseline information.

The information on institutions may be collected at the individual level within household surveys by asking respondents about their participation in these various institutions. Information may also be collected at the institutional level, by identifying who participates in different roles and activities. Moreover, it is important to identify whether there are rules or norms within the institutions that affect who participates.

3.3 Measuring empowerment and agency in agriculture

The increased interest in measuring women's empowerment both generally and within agriculture is motivated by the recognition that women's empowerment and gender equality are important development goals in themselves. Women's empowerment is also correlated with many development outcomes, such as such as poverty reduction, reducing hunger and malnutrition, and attaining good health and well-being for women and children (Cunningham, Ruel, Ferguson, & Uauy, 2015; Heckert, Olney, & Ruel, 2019; Malapit, Kadiyala, Quisumbing, Cunningham, & Tyagi, 2015; Sraboni, Malapit, Quisumbing, & Ahmed, 2014).

In this section we define empowerment, briefly review approaches to measuring it, and discuss metrics in current use. We identify both the benefits and the challenges of existing measures. We focus on quantitative measures of empowerment, but note that qualitative methods of understanding empowerment have a rich tradition of scholarship. Indeed, many of the quantitative measures discussed below have been informed by qualitative work and would ideally be implemented alongside qualitative protocols. A review of both qualitative and quantitative methods of measuring empowerment in agriculture can be found in Elias et al. (2021). Since time use features prominently in recent measures of empowerment, we also briefly review recent developments in measuring time use.

3.3.1 Defining empowerment[6]

Kabeer (1999) defines empowerment as the process by which people expand their ability to make strategic life choices, particularly in contexts in which this ability had been denied to them. The ability to exercise choice encompasses three dimensions: resources (including not only access but also future claims to material, human, and social resources), agency (the ability to define one's goals and to act on them, including processes of decision-making, negotiation, and even deception and manipulation), and achievements (well-being outcomes). Because measures of resources and achievements are relatively well developed, recent efforts have focused on measuring agency. Agency can be linked to a typology of power as it relates to gender and empowerment (Rowlands, 1995). This typology juxtaposes the notion of dominating or exerting "power over" others, with generative forms of empowerment, including "power within" (involving self-respect, self-efficacy, and an awareness of rights), "power to" (enacting personal goals), and "power with" (acting collectively toward shared interests). Framing this in terms of agency translates "power within," "power to," and "power with" to intrinsic, instrumental, and collective agency, the measurement of which is a goal of a growing body of research.

In their paper on approaches toward measuring women's agency, Donald et al. (2020) propose a multidimensional framework for capturing individual agency and review existing measurement approaches and their relative quality. They highlight the links between the psychology and the economics literature through the

[6]This section draws on Alkire et al. (2013) and Malapit et al. (2019).

association of agency with autonomy (in psychology) and with bargaining power (in economics). Indeed, some measures of autonomy, such as the relative autonomy index (RAI) are part of recently developed women's empowerment indices.

Existing measures are reported at the national level because they rely on administrative or aggregate data, and thus are measures of gender equality, rather than women's empowerment. Some examples are the Gender Gap Index (World Economic Forum (2018) and prior years), Gender Development Index (GDI), and Gender Inequality Index (GII) (UNDP, 2018). These indices measure gender inequalities in a broad set of domains but do not measure women's empowerment comprehensively or rely on only indirect proxies, such as women's age, schooling attainment, and share of parliamentary seats. Moreover, because these indices rely on national-level aggregate data, they cannot be decomposed by region or population subgroups.

In contrast, other efforts to measure women's empowerment using individual data from nationally representative surveys such as some Demographic and Health Surveys (DHS), enable researchers to measure women's empowerment, but not gender equality, because the questions are typically asked only of women, precluding comparisons with men. Moreover, the DHS include a range of questions about decision-making within the household, but these are typically confined to the domestic sphere and do not encompass decisions in the productive and economic spheres. Nevertheless, DHS data can be used in innovative ways to measure empowerment. For example, Annan, Donald, Goldstein, Martinez, and Koolwal (2021) analyze DHS data from 23 Sub-Saharan African countries and find that women "taking power" (declaring higher decision-making ability for themselves than their husbands report) vs. being "given power" (husband reports higher decision-making for his wife than she does for herself) is better for her reproductive health and her children's health.

3.3.2 The Women's Empowerment in Agriculture Index (WEAI) and its variants

The WEAI, which was launched in 2012, is a survey-based index that measures women's empowerment in the agricultural sector using individual-level data collected from male and female household members. The WEAI is an aggregate index, reported at the country or subnational level and comprised of two subindices. The first subindex assesses the degree to which respondents are empowered in five domains of empowerment (5DE) in agriculture, namely, decisions about agricultural production, access to and decision-making power about productive resources, control of use of income, leadership in the community, and time allocation (Alkire et al., 2013). The 5DE are measured using 10 indicators with their corresponding weights. Each person is given a binary score in each of the 10 indicators, reflecting whether or not she has adequate achievement. An individual level empowerment score is then generated, in which the weights of those indicators in which the individual enjoys adequacy are summed to create a score that lies between 0% and 100%. All in all, a person is defined as empowered in 5DE if she or he has adequate achievements

in four of the five domains or is empowered in some combination of the weighted indicators that reflect 80% total adequacy or more. The second subindex, the Gender Parity Index (GPI), reflects the percentage of women who are empowered or whose achievements are at least as high as the men in their households. For those households that have not achieved gender parity, the GPI shows the gap that needs to be closed for women to reach the same level of empowerment as men in their households (Alkire et al., 2013). The WEAI is the weighted sum of 5DE and GPI, which are weighted at 0.90 and 0.10, respectively.

As one of the Alkire-Foster (Alkire & Foster, 2011) family of indices and as an aggregate, headline figure, the WEAI is decomposable both at multiple levels depending on the data's sampling design and into its component subindices and indicators. Because the WEAI uses data from both male and female respondents, one can make direct comparisons between men and women in the same household and separately diagnose the aggregate sources of disempowerment for men and women. The transparency of the WEAI stems directly from its counting-based measurement approach, for which the definitions, thresholds, and weights of each indicator are explicitly defined (Alkire et al., 2013).

The WEAI has since been modified in response to needs expressed by its users. The desire to reduce survey administration time (and field costs) led to the development of the Abbreviated-WEAI (A-WEAI), which had the same five domains, but only six indicators (Malapit, Sproule, & Kovarik, 2017). The most recent adaptation of the WEAI is the project-level WEAI (pro-WEAI), developed in collaboration with 13 agricultural development projects (Malapit et al., 2019) in response to their demand for a metric that captured aspects of empowerment that they deemed essential for project success. Similar to the original WEAI, the pro-WEAI is an additive and decomposable index based on the Alkire-Foster methodology (Malapit et al., 2019). Pro-WEAI is based on a weighted adequacy count across 12 indicators that measure intrinsic agency, instrumental agency, and collective agency, which are conceptually aligned with Rowlands' (1995) "power within," "power to," and "power with." For each of these indicators, individuals are classified as adequate or inadequate based on predetermined thresholds. The pro-WEAI is composed of the 3DE subindex (three domains of empowerment, the pro-WEAI analog of the 5DE in the WEAI) and the GPI, which are weighted similarly in the pro-WEAI as the original WEAI, at 0.9 and 0.1, respectively.

3.3.3 The Women's Empowerment in Livestock Index (WELI)

The Women's Empowerment in Livestock Index (WELI) was developed to address the need for an empowerment measure that was more appropriate to settings where livestock farming is the dominant form of livelihood (Galiè, Teufel, Baltenweck, & Girard, 2019). The WELI contains six dimensions of empowerment: (1) decisions about agricultural production; (2) decisions related to nutrition; (3) access to and control over resources; (4) control and use of income; (5) access to and control of opportunities; and (6) workload and control over own time. Similar to the WEAI, each dimension was operationalized with two to three indicators, and the indicators

were used to classify a woman as adequate or inadequate with respect to the indicator. Adequacy was assessed across a range of decisions related to livestock farming and nutrition dimensions. In aggregating the indicators, each dimension was weighted 1/6 and the indicators within each dimension were weighted equally.

3.3.4 Other approaches to measuring empowerment

While not exclusively focused on agriculture, two recent metrics illustrate different approaches to measuring empowerment. The Women's Empowerment in Nutrition Index (WENI) (Narayanan, Lentz, Fontana, De, & Kulkarni, 2019) was developed to measure empowerment in nutrition, defined as the process by which individuals acquire the capacity to be well fed and healthy, in a context where this capacity was previously denied to them. This process entails acquiring knowledge about, and a say over, nutritional and health practices; gaining access to and control over intake of adequate and nutritious food; and being able to draw support from both family and other institutions to secure and maintain an adequate diet and health. Unlike the WEAI, it does not focus on agriculture and could potentially be used to examine nutritional empowerment in nonagricultural settings. It is a multidimensional index that comprises 40 indicators spanning the domains of food, health and institutions (50 indicators when expanded to include the domain of fertility for younger women) that drive nutrition. These indicators were identified from a larger set of 128 candidate indicators through a combination of normative and statistical assessments; these are then aggregated using the Alkire-Foster approach to constructing multidimensional indices. Narayanan et al. (2019) apply WENI to data from two sites in India and show that it consistently predicts nutritional status.

Other approaches for developing measures of women's empowerment have included exploratory factor analysis, and more recently, applications of item response theory (IRT) and structural equation modeling. Such methods are especially useful for identifying survey questions that are valid measures of multifaceted constructs, like women's agency. To be valid, such measures need to be conceptually sound and empirically (or psychometrically) "comparable" across groups and over time. An application of this approach is illustrated by the development and validation of the Women's Agency Scale 61 (WAS-61) among married women in Matlab, Bangladesh. Using theory, qualitative research, and questionnaire reviews, Yount, James-Hawkins, and Abdul Rahim (2020) developed an item pool to capture women's intrinsic, instrumental, and collective agency. The authors performed exploratory then confirmatory factor analyses (EFA/CFA) in random split-half samples, and estimated pairwise and factor score correlations to confirm that the dimensions of agency were distinct but related in expected ways. The authors find that women's intrinsic agency in voice/mobility outside the home is strongly related to their instrumental and collective agency, and women's instrumental agency in use of financial services and voice/mobility outside the home is strongly related to their collective agency. These findings suggest that all dimensions of women's conscientization are interrelated and necessary for women's empowerment and well-being. Quantitative approaches to measuring empowerment, particularly index-based measures, have been criticized for failing to capture the views of respondents themselves.

Some indicators may indicate empowerment in some context but not in others, and normative indicators may reflect the values of those measuring rather than those whose empowerment is assessed (Kabeer, 1999). In a study that used both the WEAI and qualitative work in Nepal, O'Hara and Clement (2018) point out the limitations of assessing empowerment using visible measures of agency, suggesting that including other forms of power (such as "power within" (Rowlands, 1995)) may be more relevant. They develop an index of critical consciousness based on domains representative of gender norms in Nepal (gender-based violence, beating, obedience, politics, chores, education for girls) and find that these are better able to capture local meanings of empowerment than the WEAI. We reiterate that a full understanding of such a complex, context-specific concept as empowerment would do well to use both quantitative and qualitative methods.

3.4 Measuring time use

Control over the use of one's time is central to women's agency. Gender disparities in time spent on unpaid domestic and care work exist in both low- and higher-income countries. On average, women spend at least three times as many hours as men on unpaid work and have a higher total work burden than men (United Nations, 2015). Yet, gender disparities in unpaid work often go undiagnosed by traditional household surveys. Time-use surveys, on the other hand, are well-suited for measuring unpaid work. Accounting for multitasking is a particular challenge in collecting data on women's time use, as child care is often done simultaneously with other activities. Lentz, Bezner Kerr, Patel, Dakishoni, and Lupafya (2019) for example, argues that the women themselves often don't recognize much of the time they spend caring for children.

Seymour, Malapit, and Quisumbing (2020) analyze differences between time-use estimates obtained from 7-day recall stylized questions and 24-h recall time diaries using data from WEAI surveys in Bangladesh and Uganda in which respondents answered both types of questions. They find stylized questions do not always produce shorter interviews compared to time diaries because of the greater cognitive burden placed on respondents by stylized questions, which require respondents to mentally aggregate activities over the reference period. In contrast, the sequential nature of the time diary, coupled with a shorter recall period, may improve respondents' memory information processing and lead to shorter interviews. The authors also find systematic differences in time-use estimates linked to the type of activity, respondent characteristics, and enumerator fixed effects for men and women in both countries. The latter-most underscores the importance of having well-trained staff to implement the time-use interviews.

Time diaries, however, do not measure the energy expenditure associated with each activity. Wearable accelerometry devices (Zanello, Srinivasan, & Nkegbe, 2017) can be used to capture energy expenditure and time use. A pilot study in Ghana using accelerometry devices, a household survey, and daily interviews, including a 24-h recall of sequential time allocation for different activities and individual food consumption, confirms the stylized facts relating to energy expenditure of men and women in rural households (e.g., that energy expenditure for men is greater than

women, men undertake the more energy intensive agricultural production tasks) and also provide insights into the physical exertion and time demands of agricultural livelihoods. Because women are required to maintain a higher physical activity level through the day compared to men, there may be constraints to labor intensification for women through the adoption of new technologies or involvement in additional economic activities.

Another approach takes advantage of a pictorial smartphone application that allows real-time recording of time-use, which eliminates recall bias (Daum, Capezzone, & Birner, 2020). The smartphone application, Timetracker, allows the self-recording of data by respondents and is based on visual tools to ensure that respondents with low levels of literacy (typically women, in many contexts) and children can record data. The proof-of-concept study, conducted in the Eastern Province, Zambia, differentiated respondents according to method of land preparation they used (manual labor, animal power, mechanical traction). The study findings suggest a strong gender differentiation for land preparation activities only among households using mechanical traction and provides evidence that the surplus time associated with mechanization is used for off-farm and domestic work.

Another phone based approach was used by Rost (2020), who followed up a time use survey with spot phone calls, asking respondents what they were doing at that particular moment. This approach did not require that respondents have smart phones. It provides a cost-effective means for data collection across seasons and finds higher levels of responses that are typically undercounted, such as people responding that they were "doing nothing" or men's participation in unpaid care and domestic work.

4 Gender and agricultural production

Gender and gender relations influence agricultural production in many different ways. Men and women may do different tasks on a farm, they may provide different amounts of labor, they may be responsible for different crops, or they may manage their own plots. They may have different access to the resources needed for farming as well as to marketing channels for their outputs. They may have different other responsibilities within the household and the community that affect the time they spend working in agriculture. As noted above, collecting information both about and from both men and women on the range of activities in which they are engaged, both in terms of labor and decision-making, will provide a more accurate understanding of farm enterprises.

The choices of which crops to grow may be influenced by the gender of the farmer. While a strict division of crops by gender is uncommon, there are often patterns whereby some crops are more likely to be grown by women while others by men.[7]

[7]See Doss (2002) for a discussion of this in Ghana.

These may be due to social norms about what is appropriate for men and women to grow, by the gendered access to both input and output markets for particular crops, or by preferences regarding the timing of labor inputs and the stream of outputs. For example, Njuki et al. (2014) find that women in Tanzania and Kenya prefer to grow leafy vegetables that can be sold on a weekly basis for a steady stream of cash to meet daily needs than to grow crops that are harvested and sold at one time. The patterns of who grows which crops are subject to change over the course of rural transformation. As crops commercialize, men may take over their production and marketing.

4.1 Access to resources for agricultural production

The evidence is widespread that in almost all contexts, women have more limited access to the resources needed for agricultural production than do men. These resources include land, cash and credit, information and extension services. In addition, gender influences the extent to which individuals can access markets, whether for inputs or for selling outputs.

Until relatively recently, there was little nationally representative sex-disaggregated data on land ownership. The questions about land ownership were typically asked at the level of the household, rather than the individual. But increasingly such data is available for Latin America (Deere & Leon, 2003), Africa (Doss, Kovarik, Peterman, Quisumbing, & van den Bold, 2015) and Asia (Kieran, Sproule, Doss, Quisumbing, & Kim, 2015). These data clearly demonstrate that women are less likely to own land, own smaller plots of land, and own land with lower value and quality than men.

The Gender and Land Rights Database[8] compiles information from national statistical agencies on gender and land rights. In only 3 of the 104 countries for which there is data, are women close to half of the agricultural holders. These countries are Cape Verde, Lithuania, and Latvia. In 72% of the countries, women are fewer than 25% of the agricultural holders. The holder is the person with the economic rights to the holding, typically meaning that they are the one who controls the production processes and outputs. Since women typically have smaller plots than men, the share of agricultural land held by women would be smaller.

Interventions have sought to improve either women's land rights or women's tenure security. Land titling programs, including those in Rwanda (Ali, Deininger, & Goldstein, 2014) and Ethiopia[9] have explicitly worked to ensure that women's names are included on the titles as owners. In several countries in Latin America, the double signature laws require that both the husband and wife must sign documents selling land. A wide range of other interventions include a community-based legal aid

[8]http://www.fao.org/gender-landrights-database/data-map/statistics/en/?sta_id=982.
[9]See Deininger, Ali, Holden, and Zevenbergen (2008) on Ethiopia's first stage land certification; Persha, Greif, and Huntington (2017) on Ethiopia's second-level certification.

program in Tanzania (Mueller, Billings, Mogues, Peterman, & Wineman, 2018) and a program of microland distribution in West Bengal, India (Santos, Fletschner, Savath, & Peterman, 2014).

While the literature on agricultural credit is extensive, relatively little work focuses on women's access to rural financial markets. Fletschner and Kenney (2014) identify key challenges that rural women face in accessing financial services, including lack of property rights, cultural norms and family responsibilities, and institutional discrimination. While microcredit is widely targeted at women, many of the microcredit programs are not designed to support investments in agriculture. For example, the repayment schedules are rarely synchronized with the agricultural seasons.

Another source of income for agriculture investments often comes from remittances. When men migrate out of rural areas, they often send remittances home. The money may be under the control of the women who are managing the farms. Yet remittances may take a long time to materialize while migrants pay off the costs of the migration and get set up in new locations; some migrants never send home any money. The remittances may be used to hire labor to replace those who have migrated or may be used to invest in land or other inputs.

It is widely documented that gender plays a role in terms of who receives information and extension services, with women being less likely to receive such services (Peterman, Behrman, & Quisumbing, 2014). While an extensive literature explores the best way to provide extension services and evaluates a range of programs, relatively little of it focuses on the role that gender plays.

One set of studies explores ways to reach more women with extension service. It is worth noting that this literature is not new; in 1990 Saito and Weidemann wrote a review paper that discussed the best practices in reaching African women farmers with extension services (Saito & Weidemann, 1990). More recent work has evaluated interventions designed to reach women farmers. For example, a randomized control trial in Mozambique suggests that women farmers were more likely to learn about agricultural techniques, particularly pit planting, in communities in which there was a woman messenger or contact farmer, in addition to a male one (Kondylis, Mueller, Sheriff, & Zhu, 2016). Similarly, a program in Ethiopia found that when extension agents were given gender training and taught to work with women farmers, the regularity of contact with extension agents increase by approximately 10% points for women, both those living in male-headed households and those living in female headed households (Buehren, Goldstein, Molina, & Vaillant, 2019). Married women may not have their own plots of land, and thus, it is useful to study the impact of providing extension to women on overall household productivity.

A set of decomposition analyses of the productivity of plots managed by men and by women finds that in Ethiopia and Uganda, the returns to extension services are lower for women farmers than for men farmers (O'Sullivan, Rao, Banerjee, Gulati, & Vinez, 2014). This may be because the extension training is less suited for the conditions under which women are farming or that the quality of the training that they receive is worse than that received by men.

4.2 Agricultural production and productivity

Many analyses of men's and women's agricultural productivity consider the "household farm enterprise" as the production unit and compare the productivity of different households, defining households as "male" or "female," typically by making a distinction between households headed by men and women. While this approach is relatively simple, it ignores the contributions that women make to farms in households headed by men (and conversely contributions that men make to farms in households headed by women). This not only ignores the joint structure of households headed by couples; but it also introduces a possibly spurious comparison between households that differ in many other ways. This approach was widely used in early studies, when little data was available about men's and women's plots and continues to be used in areas with household production systems. More recent approaches may consider men's and women's roles in decision-making as well as measures of women's empowerment in agriculture.

With the increased availability of plot-level data, an approach that is becoming widely used, particularly regarding African agriculture, is to compare the output on men's and women's plots. There are challenges, both conceptual and empirical, to assigning plots to men and women. Typically, output from a plot is assigned to an individual farmer, defined either as the person who "manages" the plot or the person who "controls output" from the plot. The latter is often referred to as the economic owner. The output from plots managed by women is then treated as women's production. In contexts where multiple plots within the household farm enterprise are managed by different people, this can be a useful approach. It does not directly account for any contributions that men make on women's plots and vice versa.

Most large-scale surveys typically find large differences in yield across plots of land, even within geographically narrow areas. These may be due differences in plot-level characteristics, such as soil quality or availability of irrigation. Or plots may experience different shocks within a growing season, such as weather or exposure to pests or disease. Gollin and Udry (2021) find both significant yield differences as well as input use intensity across plots farmed by a single farmer, growing the same crops within the same growing season. Plots may be differentially located with respect to infrastructure, including roads and markets. Other differences may result from the choices of farmers, such as which variety of seed to use, whether to use fertilizer, and particular management practices. Finally, the characteristics of the farmer may be important, such as farming experience or level of education. The gender of the farmer may be correlated with all, some, or none of these factors.

This means that comparisons of output for men and women farmers are difficult both to make and to interpret. Where men and women own and/or manage separate plots, yield differences could be due to the managerial abilities of the men and women farmers or to their physical strength or other attributes. But the differences, especially when output is measured as revenue or profit, may also be due to plot characteristics or other to the many complicated choices that households make about how

to allocate resources and prioritize investments across a range of agricultural and nonagricultural activities. Thus, a comparison of yields on male- and female-owned plots, or yields attained by male- and female-headed households, has little or nothing to tell us about the potential returns to investments in agriculture that target women.

Despite these challenges, a substantial body of empirical work has sought to compare men's and women's agricultural productivity. Almost all of these papers use observational data. This literature on gender differentials in agricultural productivity can be broadly categorized as using three approaches: (1) estimating production and profit functions; (2) applying Kitagawa-Oaxaca-Blinder decomposition to calculate endowment and structural effects on productivity differentials, and (3) comparing men's and women's labor productivity in agriculture. Some research uses technology adoption as a proxy for productivity, but technology adoption is discussed in a separate section.

4.2.1 Estimating plot level production and profit functions

In this approach, regression analysis is used with observational data to estimate the contribution of various inputs to the attained output. This lets us move beyond a simple comparison of men's and women's average output to ask about men's and women's levels of output, measured as yield, revenue, or profit.

The intuition behind these regressions is that they establish whether there is a significant relationship between the sex dummy variable and output after controlling for observable input differences and other plot or farmer characteristics. Although there are numerous studies and many conflicting results, it is fair to say that in general this literature finds little or no significant difference in men's and women's agricultural productivity after controlling for access to inputs, plot, farmer, and household characteristics. This is usually interpreted to mean that women would be equally productive as men if they were given the same access to resources.

One of the earliest analyses of productivity differences found statistically insignificant differences between men and women maize plot managers in Kenya, after accounting for input use, farmer characteristics and access to extension service (Moock, 1976). A number of other early studies use small-sample farm surveys to analyze these productivity differences, using Cobb-Douglas production functions (Adeleke, Fabiyi, Ajiboye, & Matanmi, 2008; Saito, Mekonnen, & Spurling, 1994).

In some regions, such as parts of the Sahel, some plots are farmed individually, by men or women, while other plots are farmed collectively by household members. It can be useful to categorize plots not simply as men's and women's plots, but also to have a category for joint plots or collective household plots. Guirkinger, Platteau, and Goetghebuer (2015) find that in Mali, even after controlling for crop and household fixed effects, the value of the crops grown on women's individual plots and on collective family plots is lower than on men's individual plots. This contrasts somewhat with the findings of Kazianga and Wahhaj (2013), who don't find a gender gap between men's and women's individual plots but find that household plots, which are owned by the household head and farmed collectively, are more productive.

Most of these analyses compare the productivity differences of men and women across the entire sample. In other words, they treat each plot as a unit of observation, noting the gender of the manager. In an analysis of productivity differences between men and women in the same households, Udry (1996) finds that these intrahousehold differences in productivity across men's and women's plots in Burkina Faso were large and could be attributed to differences in the intensity with which labor and fertilizer were applied to these plots. This is one of the few studies that considers the impact of reallocating inputs, rather than providing women with an amount similar to that of men (which would be a net increase in the levels of fertilizer used overall).

Much less work explicitly considers cash crops, in part because the assumption is often made that women play a relatively small role in cash crop production. A study on cocoa farmers in Ghana (Hill & Vigneri, 2014) finds no gender gap in productivity after controlling for resource use (labor, fertilizer, insecticides, farm equipment; rainfall and farm quality).

4.2.2 Extensions of modeling production functions

The more recent literature wrestles with underlying methodological issues. Many of the studies above only consider one crop rather than the portfolio of crops that famers grow across plots. Women may obtain lower values for their agricultural production because of the mix of crops that they produce. Even if men and women are equally productive in producing each crop, if women systematically choose to produce more of the lower value crops (or if households assign them to this role), their overall revenue and profit will be lower. One approach is to model both the crop choice and the level of output. Two analyses that explicitly model endogenous crop choice and productivity differentials between men and women farmers find that although the value of the crops grown by women is lower those grown by men, the disparity primarily reflects differences in the crops grown (Peterman, Quisumbing, & Nkonya, 2011; wa Gĩthĩnji, Konstantinidis, & Barenberg, 2014).

These estimations assume that inputs are exogenously determined, rather than considering that the use of such inputs is a choice, albeit a constrained choice. But particularly in analyses of gender differences in productivity, this assumption may not be appropriate. Men and women may use different amounts of fertilizer because their other labor commitments differ, affecting the time available for applying fertilizer. Alternatively, their fertilizer use may reflect differential access to markets or differential ability to mobilize cash or credit to make purchases. Thus, gender may condition whether or not one uses inputs.

Overall, the literature applying production or profit functions finds little or no significant difference in men's and women's agricultural productivity or technical efficiency after controlling for access to inputs and for characteristics of plots, households and farmers. This body of evidence has frequently been used to support a claim that women are as productive as men in farming—and that any gender differences in productivity are due to disparities in access to resources. This evidence is used to support the claim that policy should increase women's access to resources. Although increasing the amount of resources available to farmers should increase overall

productivity, the evidence doesn't necessarily suggest that the best way to allocate an additional unit of inputs is to provide it to women rather than men. A number of assumptions, many of which may be valid, are needed to support this claim.

4.2.3 Decomposition of productivity differentials

Decomposition analyses provide a means to identify how much of the productivity gap is due to this differential access to productive inputs, including land, labor, seeds, fertilizer and information, and how much is due to the differential returns to these inputs.

The most frequently used technique is a Kitagawa-Oaxaca-Blinder (KOB) decomposition. This distinguishes between (i) the endowment effect or the differences in average characteristics of productivity generating factors, such as land, labor and other inputs; and (ii) the structural effects, which are gender differences in the returns to these factors. A recent set of papers use the value of total crop output per hectare as the measure of productivity and compares plots managed by men with those managed by women.[10] Analyses from six countries (Ethiopia, Malawi, Niger, Nigeria (separate estimates for the North and South), Tanzania, and Uganda) find gender gaps in productivity when simply comparing the differences in value of output. Controlling for plot size and geographic factors, the gender differences increase in six of the seven estimations, often dramatically.

One of the reasons for these differences is related to farm size. For Malawi, Niger, Nigeria (North and South), and Uganda, after accounting for the differences in farm size, the gender gap widens. Women, on average, have smaller holdings than men. In these countries, among smallholder farms, there is an inverse relationship between farm size and productivity; smaller farmers are more productive per unit of land. Thus, simulating what would happen if women had larger farm sizes, similar to those of men, they find that the gender gaps would increase. This finding has the perverse implication that the productivity of land could be increased by dividing farms and making them smaller. It underscores that a focus on land productivity obscures the fact that we may care much more about the productivity of labor, which is more closely linked to the well-being of the poor.

The main objective of these studies is to move beyond quantifying the gender gap to identify the contribution of each factor to the gender gap in productivity. The endowment effect is significant and in men's favor for Malawi and south Nigeria, while in north Nigeria, Uganda and Tanzania, it is in women's favor. In Ethiopia, both endowment and structural effects are insignificant. Significant positive structural effects are found in Malawi, Niger, northern Nigeria, Uganda and Tanzania, indicating that most of the gap is due to gender differences in returns to factors.

The individual country analyses provide more detail. In Uganda, because so few purchased inputs are used (fertilizer, chemicals, and improved seed) by any farmers,

[10]These results are summarized in Leveling the Field (World Bank, ONE Campaign, 2014) which is based on many of the papers cited in this section. The comparative analysis here is drawn from the report. In addition, many of the individual papers are cited for their country-specific contributions.

women's lack of access to these does not contribute much to the gender productivity gap (Ali, Bowen, Deininger, & Duponchel, 2015). In Nigeria, there are significant differences in gender productivity in the North and in the South. In the North, not only do women have less access to inputs, but the returns to these factors of production are also lower (Oseni, Corral, Goldstein, & Winters, 2015).

To understand the patterns of gender differences it is useful to have information not simply on a median household or an average household, but across the entire productivity distribution. The results overall suggest that the gender gap in productivity is highest above the middle of productivity distribution. For Ethiopia, access to resources matters more at the upper end of the productivity distribution (Aguilar, Carranza, Goldstein, Kilic, & Oseni, 2015). At lower end of the distribution, most of the gap is explained by endowment effect, which implies that at lower levels of productivity, men and women have similar returns to factors and residual gender gaps are driven by women's lower access to inputs. A different explanation is offered by Slavchevska (2015, p. 349), "at the lower end of the productivity distribution, women command lower returns to the factors of production, but their smaller plots mask the gender gap in returns." Moving up the productivity distribution, structural effects become more important than endowment effects. Distance to plot or distance to nearest market, returns to labor use and child dependency ratio are the consistent factors contributing to female structural disadvantage.

While most of the decomposition studies simply compare plots managed by men and women, de la O Campos, Covarrubias, and Patron (2016) use a decomposition analysis that also considers jointly managed plots. They find that the productivity gaps are higher between male-only managed and female-only managed plots than between female-only managed plots and those jointly managed.

4.2.4 Analyzing labor productivity

A related set of studies compares the labor productivity of men and women in agriculture. This approach uses detailed information on labor inputs to identify how they are related to output levels. This approach does not require that output be disaggregated by gender, allowing us to analyze how the labor inputs of men and women affect total farm productivity.

Measuring labor inputs presents many challenges, even before we begin to disaggregate by gender, as discussed above and in the Chapter "Agricultural data collection to minimize measurement error and maximize coverage" by Carletto et al. It implicitly assumes that an hour or day of work is a useful measure of input and that the contribution of each unit is equal.

Men's and women's labor may not be perfect substitutes. For a variety of reasons, including social norms, skills, and physical capabilities, there may be differences in the labor provided by men and women. In addition, because women typically have much higher burdens in terms of the time and physical requirements of providing care labor—such as caring for children, preparing meals, fetching water, and doing

laundry—they have less time available for agricultural work. The opportunity costs of labor for men and women may also differ, since the nonfarm income-earning opportunities income may be fewer for women.

An additional challenge in estimating the contributions of men's labor and women's labor separately is that the usual measure of labor input—time spent working on a farm or plot—does not necessarily account for knowledge or management skills. A family member might work few hours but provide direction to others; this will not be accounted for in the data.

The usual approach is to treat men's labor and women's labor as separate inputs in an agricultural production function. To determine the marginal contribution to output, the indirect estimates of production elasticities of different inputs are derived. In simple terms, household output value (or some other measure of production) is regressed on a set of inputs, including men's labor and women's labor. The elasticities can be compared; it is straightforward to test statistically the hypothesis that the two elasticities are identical.

Results can be quite sensitive to the specification of the functional form. Most studies that estimate the distinct labor contributions of men and women use a translog production function, estimated from household-level data on output and inputs, including men's labor and women's labor. This estimation gives the marginal product of factor inputs, including labor. While a study of rice farmers in Nepal (Aly & Shields, 2010) finds that there is no gender gap in labor productivity, earlier work using a similar approach, finds that the marginal productivity of labor varies across seasons and crops (Kumar & Hotchkiss, 1988). They attribute these differences to women's other responsibilities across seasons. Other studies find that the marginal productivity of men's and women's labor differs by activity, such as cropping or livestock (Jacoby, 1992) and by crop (Laufer, 1985). Consistent with the gender division of labor, men's and women's labor are complements rather than substitutes.

Because men and women typically specialize fully or partially in different types of farm labor, it is difficult to interpret these findings. If we know that on the margin, an hour of women's time spent weeding contributes less to overall productivity that an additional hour of men's time spent in preparing the land, how should we direct interventions? If the marginal product of women's labor is high, this could imply that it would be useful to free up some additional time for women to contribute to agriculture. But it is equally true that if the marginal product of women's labor is low, it might support an argument that there are particularly high payoffs to redressing this imbalance.

A variation on this approach is to consider not only men's and women's labor, but also their levels of empowerment. Seymour (2017) includes a measure of the within household gender gap based on the Women's Empowerment in Agriculture Index. He estimates a stochastic frontier production function model and finds that lower empowerment gaps between spouses are associated with higher levels of technical efficiency. This result holds both for plots that women jointly manage with their spouses, as well as those that women do not actively manage.

The studies of agricultural productivity discussed thus far all rely on observational data. Few studies evaluate the impact of interventions on the gender gaps in agricultural productivity. Two analyses consider the actual provision of inputs to men and women in Malawi, rather than estimating what women's productivity would be if they used the same inputs as men (Gilbert, Sakala, & Benson, 2002; Karamba & Winters, 2015); neither finds a gender gap in productivity.

4.3 Technology adoption

The use of improved technologies, including improved seeds, fertilizer, and some management technologies, is often used as an indicator of productivity in analyses of agricultural productivity. The implicit assumption is that farmers will not adopt these technologies if they do not increase productivity, although a more appropriate assumption is that farmers will not adopt them unless they increase profit. This approach does not require detailed information on the use of inputs and outputs, nor that a value for labor inputs be calculated; instead, researchers assume that farmers take the labor requirements into consideration when deciding whether or not to adopt.

An extensive literature uses observational data and attempts to understand which farmers use improved technologies and why more farmers do not. Much of this literature still only compares the adoption patterns of male- and female-headed households. Better approaches are to compare the use of technologies on plots managed by women with those managed by men or to consider how factors within the household, such as who makes decisions, how risk preferences vary within the household, or levels of women's empowerment, affect technology adoption outcomes.

In their simplest form, these studies use regression analyses to identify the characteristics of adopters. Including a variable for the sex of the farmer provides a simple way to see if the patterns differ for men and women. Several studies suggest that, all else equal, men and women are equally likely to adopt these technologies. The challenge is that it is rare that all else is equal.

The adoption patterns may vary across technology types. Analyzing the adoption of a set of sustainable intensification technologies, Theriault, Smale, and Haider (2017) use plot level data and control for the characteristics of the plot manager. They find that women managers are less likely than men managers to adopt yield-enhancing and soil-restoring strategies. They find no gender differences in the adoption of yield-protecting strategies. The factors influencing adoption also differ for men and women managers.

Household structure may also affect adoption. Doss and Morris (2001) find that holding other inputs constant, men and women farmers in male-headed households were indistinguishable in their adoption of improved seed and fertilizer. However, women farmers in female-headed households were less likely to use these improved technologies. This suggests that there are unobservable differences between women living in male- and female-headed households. It may be that women living in male-headed households have access to information and some unobservable resources that

are unavailable to women living in households without men. Chirwa (2005) finds that women farmers in female-headed households are less likely to adopt fertilizers but equally likely as men and women in male-headed households to adopt hybrid seeds. These studies demonstrate that if we only consider the sex of the household head, we will miss the women farmers living in male-headed households who are adopting technologies.

Comparisons across technology adoption on men's and women's plots works well in places where plots are farmed individually. It works less well in systems of household farming, where men and women work together on the same plots. In these systems, the household head is often considered the farmer, limiting analyses to a comparison by the gender of the head. But more recent work is now accounting for gender differences within the household. Gebre, Isoda, Rahut, Amekawa, and Nomura (2019) analyze the adoption of improved varieties of maize taking into account whether the decision-making on the farm is done by a man, woman, or jointly. They identify men making the decisions in female-headed households and vice versa. They find no gender differences in adoption, but the intensity of adoption is higher when men make the decisions and lower in the 36% of households where joint decision-making prevails. Similarly, Mutenje et al. (2019) consider women's role in household decision-making and the relationship with the adoption of a range of climate smart agricultural technologies. They develop an index of women's involvement in several agricultural decisions and find that this index is positively correlated with the adoption of a number of CSA technologies.

Analysis of other factors that may influence the adoption of technologies also may include a gender component. Mekonnen, Gerber, and Matz (2018) analyze how networks in Ethiopia influence the adoption of row planting. Networks that include an information link are associated with higher adoption of row planting, especially for women farmers.

The gender dimensions of technology adoption have also been analyzed through impact evaluations of a range of interventions. Many of these interventions provide extension or information support to farmers and evaluate the success based on whether or not the farmers adopt the technology. The interventions could be targeted to women either for use on their own plots or to see how it influences adoption at the household enterprise level. For example, Lambrecht, Vanlauwe, Merckx, and Maertens (2014) consider how an extension intervention affects the adoption of mineral fertilizers, row planting and improved legume varieties in the DRC. Conditional on plot, village, and farmer's characteristics, the likelihood of adoption is increased when men and women from the same household participate jointly in the extension program. The extension program also increased the adoption of mineral fertilizers and improved legumes by women in female-headed households.

While it is useful to know the extent to which men and women use various technologies and the extent to which differential technology adoption can be explained by access to land, information, and other relevant inputs, other questions about technology remain. Little effort has been made to specifically develop technologies for women and to see how this would affect their productivity and the gender gaps.

4.4 Migration and changes in labor patterns

One of the other ways in which agricultural production is changing—and an area where gender needs to be explicitly considered—is related to the movement of people within and out of agriculture—both men and women, youth and adults. With the changing labor patterns also comes shifts in the labor and responsibility of those remaining in the agricultural sector.

The first pattern is that of men moving out of agriculture, often out of rural areas. The migration may be domestic or international. The length of time may be seasonal or for an extended period. This movement out of agriculture by men is often referred to as the feminization of agriculture. These processes may result in women becoming more involved in agriculture, either by providing more labor or by taking additional decision-making responsibilities.

While the impacts of migration, especially international migration, on the labor supply of those remaining in sending communities has been widely studied, little of this work analyzes the gender implications. Much of the work on migration out of rural areas tends to focus on the impacts on aggregate labor supply (e.g., Hanson, 2007), rather than specifically on agricultural labor.

The studies that do consider the impacts on rural communities typically use observational data and account for the endogeneity of the migration decision. One analysis finds that when men migrate out in Nepal, women use decrease their labor market participation (Lokshin & Glinskaya, 2009). Mendola and Carletto (2012) find that in Albania, having a current migrant decreases women's market work and increases unpaid work, while past migration increases women's participation in self-employment and decreases unpaid work. They find no significant impacts on men's labor supply.

A few studies specifically analyze the impact on agricultural work. In Egypt, evidence from Binzel and Assaad (2011) finds that the decrease in market work corresponds with an increase in unpaid family work and agricultural subsistence work. Their findings hold for both participation in work and the number of hours worked. In China, Mu and Van de Walle (2011) analyze the impacts of men's migration to urban areas on the women who remain. They find a strong pattern of women moving out of nonagricultural wage work and family work, into agricultural activities. Finally, in Nepal, Slavchevska et al. (2020) find that when men migrate out, the men remaining in rural communities reduce their labor in nonfarm work without significantly increasing their labor allocation to other activities. Women, on the other hand, increase their work on the farm, taking on new responsibilities and moving from being identified as contributing family workers to primary farmers.

A second change is that women are taking over more managerial responsibilities as men migrate out. In China, a study by De Brauw, Huang, Zhang, and Rozelle (2013) finds that women have increased both their supply of labor to agriculture and management roles on the farms and that women-managed farms are as efficient as those managed by men. A rich qualitative literature demonstrates that this increased management role can be empowering for women as they gain more voice

in their household and community. Yet, it may also be disempowering, if they are expected to manage a marginalized farm with limited access to the necessary resources and inputs, including information and labor.

The impacts on agricultural households will vary depending on who is migrating. A study using panel data on youth migration in Ethiopia and Malawi (Mueller, Doss, & Quisumbing, 2018), finds that labor shortages occur when a child of the household head migrates, but the impacts differ based on whether the migrant is a son or a daughter. In Ethiopia, women who are either the head or spouse may begin working on the farm when a son leaves. In Malawi, when a son migrates, his brothers work more on the farm. In Malawi and when a daughter migrates, both sons and daughters who remain increase their farm labor.

There is continuing scope for further analyses of how agricultural labor patterns are changing, particularly as men migrate out. Yet, in some areas, women are the ones likely to move out and there have been few analyses of how this impacts agricultural labor. In addition, how changes in labor affect both agricultural production and people's well-being remains an unexplored area. Analysis of the second- and third-round effects of migration will require considering the effects of remittances on families and individuals who remain in rural areas.

5 Gender, markets, and value chains

Understanding gender is important to understanding how men and women participate in markets and value chains—what commodities they choose to produce for the market, the extent of their participation in labor and product markets, and the terms under which they participate. Women's relatively limited access to productive assets, technology, credit, information, and social networks disadvantage their participation in value chains of higher-value or contract farming arrangements; food safety and quality standards, sustainability standards, and inability to fulfill the volume requirements of large-scale buyers raise the entry costs for high-value markets in particular (Reardon et al., 2009, cited in Djurfeldt, Mawunyo Dzanku, & Cuthbert Isinika, 2018).

Value chains approaches can provide an entry point for analyzing the gender implications of moving beyond the farm to the market and eventually to the consumer. A value chain charts the sequence of actions and the organizational links that move a product or service from conception, through a series of steps, including production, processing, marketing, and delivery to final consumer, through to its consumption and disposal (Rubin & Manfre, 2014). To analyze gender in value chains, one needs to make explicit the different levels and categories of men's and women's participation in value chain activities (Rubin & Manfre, 2014), examining their roles at different nodes of the value chains, or comparing their roles at similar nodes in different chains.

Most of the literature on gender and market access consists of observational studies. Some studies focus on barriers smallholder women farmers face in market access, whether for domestic or export markets. Others examine the implications of agricultural commercialization or participation in specific value chains or nodes of the value chain on women's empowerment. The observational studies tend to cast a wider net than experimental approaches. Experimental approaches using RCTs tend to be confined to well-defined interventions that are amenable to randomization. These include, for example, changes in contractual terms, or training and grants designed to increase women's participation in markets.

5.1 Gender and commercialization: Observational studies from Africa

Discussions of gender and commercialization frequently distinguish between food crops and cash crops, with the former believed to be women's responsibility and the latter, men's (see Hill & Vigneri, 2014 for a discussion). The division of crops by gender is frequently explained by women being responsible for feeding the family, thus growing food or subsistence crops, and men being responsible for providing cash income, thus growing cash and export crops. While the boundaries between "men's" and "women's" crops often are blurred, social norms, which are context-specific, influence the types of crops that they can grow (Doss, 2002).

Several studies in the literature argue that women farmers are disadvantaged with respect to access to more lucrative cash crop markets. Hill and Vigneri (2014) review how gender inequalities in resources result in different levels of participation, methods of production, and modes of marketing cash crops, using a case study of two perennial cash crops, cocoa in Ghana and coffee in Uganda. They find that women cocoa farmers in Ghana face barriers in accessing input markets, particularly markets for labor and nonlabor inputs, leading them to adopt production technologies that are intensive in labor but not in modern inputs. Analyzing coffee transactions for a small panel of 300 coffee farmers in Uganda, they find that gender differences in marketing are largely explained by women marketing smaller quantities of coffee and not owning bicycles. Women have less access to marketing channels that allow value addition; in the marketing channels in which they are engaged, they do not face discrimination (Hill & Vigneri, 2014, p. 337).

However, Djurfeldt et al. (2018), in their longitudinal study in Ghana, Kenya, Malawi, Mozambique, Tanzania, and Zambia, find inconsistent evidence that women have limited access to cash crop markets. The production of cash crops is dominated by male farm managers only for sugar cane in Mozambique and Malawi, tobacco in Malawi, and cotton in Zambia. In Malawi, male dominance is due to membership rules in the tobacco marketing association that discriminate against women, while travel to centralized auction floors is impossible for many women whose mobility is restricted by domestic responsibilities and gender norms. While commercialization of food crops did not necessarily lead to male dominance in these

markets, gender differentials still exist. Even if increased agricultural commercialization benefited female farm managers, it disproportionately advantaged their male counterparts.

Studies of gender in modern supply chains (MSC) are designed to account for the specific features of the supply chain. Maertens and Swinnen (2012) examine two high-value horticulture export chains in Senegal: (1) the bean and mango supply chains, involving several exporting companies and based on both smallholder contract-farming and on vertically integrated agro-industrial production; and (2) the tomato supply chain, which is dominated by one multinational company and is completely vertically integrated. These different organizational arrangements have different gender implications. Women farmers tend to be excluded from supply chains using smallholder contracts, since these contracts are mostly signed by men. Although women may work on contracted plots controlled by men, the share of female labor is relatively small. In contrast, women make up a large share of the workforce on large farms, enabling them to earn wages, which enhances their bargaining power within their households.

A study of the welfare impacts of the cut flower export industry in Ethiopia (Getahun & Villanger, 2018) deliberately employs the collective model of the household. The authors hypothesize that employment in the cut flower industry improves women's intrahousehold bargaining power, both through higher wages and through working conditions that include unionization, tight working community with fellow women workers, and being away from home 6 days a week. The authors use a difference-in-difference (DID) model and identify effects by comparing women workers with those who also applied for jobs, but who never started working. Their results suggest that the woman's higher earnings from the job increased household consumption not only through higher household income, but also by reducing the earnings difference between spouses. Estimates of leisure demand functions also suggest that a flower industry job not only reduced the woman's leisure time but also her husband's because he took up some of his wife's domestic chores.

Another study in Uganda explores whether private sector sustainability standards can promote gender equality in the small farm sector (Meemken & Qaim, 2018). In 2015, the authors purposively selected two coffee farmer organizations in central Uganda, each of which was certified under a different sustainability standard. Noncertified farmers serve as the control group and include members and nonmembers of the two farmer organizations. The authors hypothesize that sustainability standards may contribute to women's economic empowerment (1) by improving women's access to markets and/or (2) by increasing women's control over income and wealth from cash crop production. They find that sustainability standards increase total household assets and those individually owned by female household heads. These positive effects are the result of higher coffee revenues in certified households. They find that, for wives in male-headed households, standards not only increase total asset ownership but also have strong positive effects on assets that are jointly owned by male household heads and their female spouses. Women in certified households also

tend to have greater control over cash revenues from coffee and other crops than their counterparts in noncertified households, even if women are not directly involved in marketing. Because certification involves a bundle of gender policies, the authors cannot disentangle the effect of components of the bundle. Follow-up research to improve the design of standards could focus on understanding the cost-effectiveness of specific gender measures and of the bundle implemented. We return to this issue in discussing a study by Ambler et al. (2021) below.

5.2 Empowerment and commercialization: Observational studies from Asia

Because gender distinctions by crop are less prevalent in Asia, where household members farm jointly, studies that examine empowerment implications of commercialization and market inclusion have focused on the extent of market participation or the opportunities for empowerment and gender equality within specific value chains or nodes within the value chain. Two observational studies in India and the Philippines use variants of the Women's Empowerment in Agriculture Index to investigate these issues.

Gupta, Pingali, and Pinstrup-Andersen (2017) classify households from a survey in Maharashtra, India, as landless, food-cropping, and cash-cropping based on the proportion of crops sold. Across the three different farming systems, women's empowerment is lowest in the landless group and highest in the cash-cropping group. Using regression analysis, the authors find that women's empowerment scores are significantly higher among women belonging to the cash-cropping and food-cropping households (relative to the landless households). They also find that women's empowerment in agriculture declines significantly as area of land owned by the household increases, possibly because as household landholdings increase, the probability that a greater proportion is owned by men also increases (Kieran et al., 2015).

Malapit et al. (2020) adapt the survey-based project-level Women's Empowerment in Agriculture Index (pro-WEAI) to measure women's and men's empowerment and to investigate their correlates in the abaca, coconut, seaweed, and swine value chains in the Philippines. Unlike in many other countries for which WEAI or pro-WEAI data exist, gender parity was high in all value chains, reflecting the generally gender egalitarian norms in Southeast Asia. Regressions examining the correlates of empowerment show that women engaged mainly in processing have lower empowerment scores compared to those engaged in production or trading, possibly because processing activities, particularly in the coconut sector, are low-value and time-demanding. Men who are laborers or wage earners are less empowered than those who are not, likely due to preferences for having one's own business, which may involve greater autonomy than wage work. Access to extension services is more strongly associated with men's empowerment than women's, possibly because extension services are directed toward men.

5.3 Experimental evidence on value chain interventions

One of the difficulties in ascertaining which package of value chain interventions works best is that many value chain interventions are bundled, as in the certification study discussed above. Thus, experimental studies of the gendered impacts of value chain interventions have attempted to unbundle these interventions by focusing on one aspect of the value chain (rather than the entire chain). Ambler et al. (2021) study the take-up and empowerment impacts of an intervention designed to increase women's economic empowerment among sugarcane farmers in Uganda. They partner with a large sugar company in Uganda that buys most of its sugarcane through contracts with smallholder farmers—almost all of whom are men—and implement a women's economic empowerment intervention that encourages these farmers to transfer a sugarcane contract to their wife or to register a previously uncontracted sugarcane block in her name. The consent of both spouses is required for the contract intervention. Prior to the contract intervention, Ambler et al. randomly assigned couples to be invited to a couples' workshop focusing on a balanced household and participatory decision-making. The workshop was based on a component of the Gender-Action Learning System (GALS), a larger community-based empowerment program. GALS and related programs recognize the importance of including men in women's empowerment initiatives, even catalyzing them as "champions" of women's economic empowerment. They find that lower socioeconomic status and household gender norms both predict a couple's refusal of the intervention, but that couples invited to the participatory decision-making workshop were less likely to refuse the subsequently offered empowerment intervention. Moreover, the workshop was effective at addressing sources of disadvantage that arise from household gender norms and division of labor, but was less effective at addressing refusal rates associated with socioeconomic status.

Bernard et al. (2020) study the effects of a novel contract design between a dairy processing factory in Northern Senegal and seminomadic milk producers. The regular contract between the processing company and its milk suppliers paid a fixed price per liter of milk delivered to the company and provided access to production inputs at discounted costs. A new contract feature, designed to address ongoing issues of irregular milk delivery and malnutrition in the study area, consisted of a reward for regular milk delivery in the form of a micronutrient fortified yogurt (MNFY) manufactured by the dairy company using the milk collected from its suppliers. The MNFY product was intended for the children of suppliers and was delivered daily to milk collection points mainly accessed by women, who were primarily responsible for delivering milk to these points. They find that the nutrition-based incentive increases the regularity and amount of milk delivered, although this effect is limited to the dry season, when the incentive is binding. The impact on milk deliveries is mainly driven by lower seasonal migration in the dry season and increases in women's decision-making with respect to milk production. They observe larger impacts on milk production and deliveries when women control the contract, possibly because these households may have different preferences than households where

men control it. Alternatively, men and women within the same household might have different preferences, but women can only assert their preferences in households where they have sufficient influence over milk production decisions. A related study (Le Port et al., 2017) finds a significant impact of the intervention on children's hemoglobin levels, illustrating the potential for leveraging value chain services for good nutrition.

6 Gender and well-being outcomes: Food security and nutrition

Tests of household models generated a large body of empirical evidence documenting positive associations between greater control of resources by women and improved human capital outcomes for children. Most of the early studies were observational studies (see reviews in Quisumbing, 2003; Schultz, 2003; Strauss & Thomas, 1995); these in turn informed the design of large-scale programs in Latin America and elsewhere, which were evaluated using randomized controlled trials (RCTs). A growing body of evidence now exists on the impacts of cash transfers on such human capital outcomes as education, health, and nutrition (Yoong, Rabinovich, & Diepeveen, 2012).

While these studies have influenced policy recommendations to direct resources to women to improve nutritional outcomes, a recent systematic review of the role of women's empowerment in child nutrition outcomes (Santoso et al., 2019) finds that the relationship between women's empowerment and child nutrition is inconclusive. Moreover, the relationship between women's empowerment and their own nutritional status remains understudied. Most of these studies also neglect the role of men and issues of gender equality. For example, many cash transfer programs are targeted to women, without exploring the counterfactual of targeting men. Neither do many studies explicitly explore relationships between gender equality and desired outcomes (Harris-Fry et al., 2020).

Ruel and Alderman (2013)'s conceptual framework identifies six pathways through which agricultural interventions can impact nutrition: (1) food access from own-production; (2) income from the sale of commodities produced; (3) food prices from changes in supply and demand; (4) women's social status and empowerment through increased access to and control over resources; (5) women's time through participation in agriculture, which can be either positive or negative for their own nutrition and that of their children; and (6) women's health and nutrition through engagement in agriculture. The last three pathways focus specifically on women, but gender dynamics can affect all six pathways. Although the conceptual framework is widely accepted, empirical evidence on the strength of these relationships remains scarce. Evidence continues to be generated using both observational studies and RCTs.

6.1 Women's empowerment in agriculture and well-being outcomes: Observational studies

The availability of a standardized measure of women's empowerment, the WEAI (Alkire et al., 2013), has greatly contributed to analyses of the role of women's empowerment in agriculture for nutrition outcomes at the household and individual levels. Two aspects of these studies are noteworthy. First, most of these are observational, thus the estimated relationships between women's empowerment and nutrition outcomes should be interpreted as associational rather than causal. Second, because the WEAI was designed to measure empowerment in agriculture, analyses have been limited to households that are engaged in agricultural activities. This limitation may wrongly misclassify women in agricultural households who are engaged in nonagricultural occupations, as disempowered. This is an important consideration, because we expect that both women and men will move out of agriculture into nonagricultural occupations as the economy transforms.

The empowerment measures and outcomes analyzed vary across studies. Sraboni et al. (2014) use the nationally representative Bangladesh Integrated Household Survey (BIHS) 2011–2012 to examine the relationship between women's empowerment in agriculture, measured using the WEAI and its component indicators, and per capita calorie availability, dietary diversity, and adult body mass index (BMI). Because empowerment itself is endogenous, they use instrumental variables (IV) regression to examine the relationship between various measures of women's empowerment and measures of household food security. They find that increases in women's in women's empowerment scores are positively associated with both calorie availability and household dietary diversity. Narrowing the gap in empowerment between men and women within households is also positively associated with calorie availability and household dietary diversity, consistent with the growing literature arguing that reducing intrahousehold gender inequality contributes positively to household welfare.

Sraboni and Quisumbing (2018), using the same data set from Bangladesh, examine associations between women's empowerment, also measured using the WEAI, with dietary quality of individuals along the life course. Their measures of dietary quality are the dietary diversity score (the number of food groups consumed) and individual intakes of macronutrients (calories and protein) and micronutrients (iron, zinc, and vitamin A), obtained from a 24-h food recall. Their findings suggest that women's empowerment is associated with better dietary quality of household members, but the strength of this association varies across the life course. Maternal schooling and household socioeconomic status are more important for younger children's dietary quality, while women's empowerment appears to be more important as a correlate of dietary diversity and nutrient intakes for adult men and women. However, women's empowerment does not benefit all individuals alike: gender bias favoring boys emerges in adolescence. These results emphasize the importance of understanding social context and gender norms in analyses of gender and agriculture. In the South Asian context, a woman investing more in sons than in daughters may be motivated by self-interest rather than altruism, given the prevailing male advantage

in labor markets and property rights, women's need for male mediation in the community, and women's dependence on sons in widowhood or old age (Agarwal, 1997).

Gupta, Pingali, and Pinstrup-Andersen (2019) examine relationships between women's empowerment, measured using the WEAI, and women's iron status in Maharashtra, India. They find that the likelihood of poor iron status in women declines as women's empowerment levels in agriculture improve; this decline is seen across a range of dietary diversity measures. This suggests that, in addition to dietary pathways, women's empowerment can play a role in addressing micronutrient deficiencies in a vulnerable subgroup of the population. These findings also reinforce the need to move away from the "staple grain fundamentalism" that has characterized agricultural policy in India, toward more nutrition-sensitive food systems.

Most of these analyses use cross-sectional data, and with few exceptions that use instrumental variables approaches (Gupta et al., 2019; Sraboni et al., 2014), do not account for the potential endogeneity of women's empowerment and its simultaneous determination with many of the outcomes under study. Another recent exception is Kassie, Fisher, Muricho, and Diiro (2020), who assess the moderating effect of women's empowerment on the relationship between agricultural technology adoption and women's dietary diversity in Kenya. They study the push-pull organic pest control system (henceforth PPT, or push-pull technology) introduced in 1998 by the International Centre of Insect Physiology and Ecology (icipe) and its partners. Kassie et al. use an endogenous switching regression approach, first estimating the probability that women belong in one of four treatments and using the estimated selection correction terms in the second stage estimation of the effect of women's empowerment status on outcomes. The four treatment categories are: (1) being a disempowered woman in a nonadopting household; (2) being an empowered woman in a nonadopting household; (3) being an empowered woman in a household that adopted PPT; and (4) being a disempowered woman in a household that adopted PPT. The authors find that women's empowerment is positively associated with the woman's dietary diversity score. Moreover, women's empowerment mediates the effect of technology adoption: while adoption of PPT has a positive nutrition effect regardless of empowerment status, its effect is stronger for households with empowered women. The above studies illustrate that standardized empowerment metrics are useful in analyzing relationships between empowerment and nutrition outcomes. However, because most studies are single-country studies, it is difficult to generalize beyond specific contexts. Pooling the data from multiple countries permits the analysis of factors that may be common across countries. Quisumbing, Sproule, Martinez, and Malapit (2021) analyze WEAI data from six countries in Africa and Asia—Bangladesh, Cambodia, Nepal, Ghana, Mozambique, and Tanzania—to identify which indicators and dimensions of women's empowerment are related to dietary and nutrition outcomes of women, children, and their households and to examine whether tradeoffs exist among these dimensions of empowerment. Using an OLS specification, the paper analyzes a common set of nutritional outcomes (household, woman, and child dietary diversity, women's BMI, and child height-for-age (HAZ) and weight-for-height (WHZ)) as a function of variables that are

found in all six data sets, using a pooled regression with country fixed effects. Few significant associations emerge between the aggregate empowerment scores and nutritional outcomes. The women's empowerment score is positively associated only with child HAZ, while lower intrahousehold inequality is associated with a higher likelihood of exclusive breastfeeding and higher HAZ but with lower BMI. Analysis of the subdomain indicators finds more significant associations, suggesting that tradeoffs exist among different dimensions of empowerment. Indicators of greater direct involvement in agricultural production, including more hours worked, are associated with worse dietary quality and lower BMI for women. The associations between BMI and work hours, particularly for poorer women in low-BMI populations, may reflect tensions among women's productive and reproductive roles, given time and energy constraints. These may also reflect cultural norms in which an empowered woman is viewed as one who can ensure her family's and children's food and nutrition security, rather than as one who takes care of her own welfare.

Shapley decompositions show that women's empowerment accounts for a small share of the variance in nutritional outcomes, with household wealth and country-level factors accounting for the largest share of the variation in household and women's dietary diversity. In contrast, most of the variation in child outcomes comes from child age. Women's empowerment is not sufficient to improve nutritional outcomes; it requires addressing the underlying determinants of poor nutrition in addition to empowering women and improving gender equality.

6.2 Gender and nutrition-sensitive agriculture: Impact evaluations

Impact evaluations of nutrition-sensitive agricultural programs (NSAP) provide opportunities for investigating how gender matters in agriculture and nutrition interventions. Nutrition-sensitive programs are those that (1) have a clearly stated objective of improving nutrition and (2) incorporate specific nutrition interventions to achieve this goal. Given that recent reviews on NSAP exist (e.g., Ruel, Quisumbing, & Balagamwala, 2018), we do not conduct an exhaustive review of the studies and their findings but focus on studies that have paid explicit attention to gender dynamics. We review four different types of NSAP: biofortification, homestead food production systems, livestock transfer programs, and value chains for nutritious foods. All of these studies use sex-disaggregated data, not only on nutrition outcomes, but also on the ownership and control of land and other assets.

Biofortification is the breeding of staple crops that are richer in essential micronutrients than traditional varieties; this has been shown to be a feasible and cost-effective approach to addressing deficiencies in vitamin A, iron, and zinc (Bouis & Saltzman, 2017). In 2007, the HarvestPlus Reaching End Users (REU) project distributed provitamin A-rich orange sweet potato (OSP) in Mozambique and Uganda to increase dietary vitamin A intakes in the local population. In order to learn about the impact and cost-effectiveness of the biofortification strategy in these two very different settings, the project incorporated cluster randomized-control trials in

each country (De Brauw, Eozenou, Gilligan, Hotz, & Kumar, 2018). The primary outcomes studied after the intervention ended were adoption, nutrition knowledge, and vitamin A intakes.

Using data from the Uganda sample, Gilligan, Kumar, McNiven, Meenakshi, and Quisumbing (2020) examine the role of women's bargaining power in the adoption and diffusion of OSP and on children's dietary intakes of vitamin A. They take advantage of detailed data not only on the identity of the plot manager, but also on multiple decision-makers on each plot. They find that the probability of adopting OSP is not affected by the exclusive or joint control of assets by women at the household level. However, within households, parcels of land under joint control, in which the woman has primary control over decision-making, are significantly more likely to contain OSP. Using data on individual and joint control of nonland assets, they find that women who control a higher share of household nonland resources are more likely to share OSP vines, showing that women use greater bargaining power (proxied by the share of nonland assets) to facilitate diffusion of this health-promoting technology. They do not find any impact of women's bargaining power on children's dietary intakes of vitamin A, possibly because husbands and wives have the same preferences regarding their children's nutritional status.

Homestead food production (HFP) programs focus on the production of nutrient-rich foods on homestead plots. These programs typically target women in smallholder agricultural households and train them to cultivate kitchen gardens and often to raise poultry or fish. Training is conducted on Village Model Farms (VMFs) and field staff train women to cultivate on their own homesteads. HFP programs are nutrition-sensitive; in addition to cultivating nutrient-rich foods, they include a behavior change communication (BCC) strategy that teaches beneficiaries optimal nutrition, health, and hygiene practices and provides them with skills to negotiate in favor of these behaviors in their household.

Evaluations of these programs have consistently found positive impacts on the diets of women and households (Girard, Self, McAuliffe, & Olude, 2012), and on the nutritional status of children and women in Burkina Faso and Cambodia, among others (Olney et al., 2016; Olney, Talukder, Iannotti, Ruel, & Quinn, 2009). In 2010, Helen Keller International (HKI) initiated a 2-year Enhanced-Homestead Food Production (E-HFP) program in Gourma Province in eastern Burkina Faso. In the program area, women farm plots independently and maintain control of the produce of these plots. The program's objectives were to improve women's production of nutrient-rich foods, their health and nutrition-related knowledge and practices, and their nutrition and health outcomes, as well as that of their children. HKI worked with communities to identify land that could be used by women for a communal farm and provided beneficiary (participating) women with agriculture inputs and chicks to start their home production activities.

HKI partnered with the International Food Policy Research Institute (IFPRI) to evaluate the program's impacts on nutrition outcomes, ownership of productive assets, and related social norms using an RCT design coupled with two rounds of qualitative research (see Olney, Pedehombga, Ruel, & Dillon, 2015; van den Bold

et al., 2015). Among villages where HKI had been operating and had access to water in the dry season, 55 villages were randomized into treatment and control arms. Intervention villages received gardening and small animal raising interventions and health and nutritional counseling through HKI's BCC strategy. Within the treatment group, in half of the villages, BCC strategy was carried out by Older Women Leaders and in the other half, it was carried out by village Health Committee members. The agricultural intervention was similar across all treatment villages.

van den Bold et al. (2015) used quantitative data from the main RCT to assess the short-term net effect of the asset transfer on intrahousehold asset distribution, and to determine whether productive assets transferred and increased income generating opportunities for women remain with the women or are reappropriated within the household. van den Bold et al. find positive impacts of the program on both the number and value of women's agricultural assets and the number of small animals owned, although men's holdings of small animals increased more and the number of their agricultural assets decreased (almost one-to-one with the increase in women's agricultural assets). This indicates a shift in the pattern of ownership which narrowed the proportional gap in ownership of agricultural assets between men and women in treatment villages; this shift did not happen in control villages. In addition, women in treatment villages also reported having control of their gardens, including the produce and income generated, although men maintained control of higher-value animals.

A follow-up study (Heckert et al., 2019) found that improvements in women's empowerment in the domains of spousal communication, purchasing decisions, healthcare decisions, and family planning decisions contributed to the program's impact on reducing wasting (low weight-for-height), with the largest share being attributable to spousal communication. Improvements in women's empowerment did not contribute to the increase in hemoglobin.

Livestock-oriented programs, many of which involve livestock transfers, have been implemented primarily as interventions to reduce poverty and improve livelihoods, and secondarily to increase farm households' production and consumption of animal-source foods (Ruel et al., 2018). Janzen, Magnan, Sharma, and Thompson (2018) evaluate the short-term (1.5 years) impacts of a livestock transfer and training program implemented by Heifer International (HI) in rural Nepal using an RCT. The program seeks to reduce poverty through self-help group (SHG) formation, livestock transfers (two doe goats), technical trainings, and values-based training. Values-based training encourages beneficiaries to "pay it forward" (PIF) by providing technical training and the first-born female offspring of their received goats to another individual in their community. Village development committees (VDCs) were randomized into three treatments and control. Janzen et al. (2018) find that the goat transfer and training program significantly increases financial inclusion and women's empowerment after 1.5 years, with similar impacts for direct beneficiaries across program variations, but impacts that are half as large in treatments without PIF. They do not find statistically significant changes in assets, incomes,

expenditures, physical health, or food security after 1.5 years. The authors attribute this to the timing of livestock growth and development, because no beneficiaries would have had marketable goats at the time of data collection.

Whether these empowerment impacts will be sustained needs further study. Janzen et al. cite the six-country evaluation of BRAC's Graduation Program (Banerjee et al., 2015), an intensive asset transfer and training program. After 3 years, the BRAC program has positive impacts on expenditure, food security, assets, finance, time use, income, and mental health. Women's empowerment initially increases but is not sustained. A critical issue is whether women maintain control of assets transferred.

A follow-up study to one of BRAC's earlier livestock transfer programs in Bangladesh, the Challenging the Frontiers of Poverty Reduction-Targeting the Ultra-Poor project, examines whether the targeted woman retained control of the transferred asset (Roy, Ara, Das, & Quisumbing, 2015). For livestock (the primary assets transferred), the program slightly increases ownership by men but causes much larger increases in sole or joint ownership by women. These increases in women's livestock ownership are associated with some increases in their control over the livestock, including the right to sell cattle, which are perceived as "men's assets" in the local context. However, Roy et al. (2015) also find increases in household ownership of many other assets (not directly transferred by the program), which tend to be solely owned by men. This suggests that proceeds from the asset transfer program, although ostensibly transferred to women, are reinvested in assets that are controlled mostly by men. Moreover, the program shifted women's work from outside to inside the home, because the transferred asset (livestock) needed to be maintained within the homestead, potentially reducing women's mobility. The program also significantly reduced women's voice in a range of decisions, consistent with economic models that link individuals' relative control over resources to their intrahousehold bargaining position. Yet, qualitative work among the women in the program suggests that there were many intangible benefits to women, such as increased confidence and social capital. Moreover, because work opportunities outside the home for poor women are poorly paid and stigmatizing given local norms of female seclusion, most beneficiary women preferred to generate income at home, even if it reduced their mobility and increased their workload. Thus, even if the program reduced their voice in household decisions, women themselves tended to frame the gains in terms of higher social status within the community and the household because of their contributions to the economic condition of the household, rather than in terms of individual gains.

Whether these nutrition-sensitive agriculture programs can also empower women is an important question. Emerging evidence from impact evaluations (Quisumbing, Meinzen-Dick, & Malapit, 2021) reveals some promising programmatic features—working through women's groups, providing agriculture, nutrition BCC, and gender sensitization training to husbands and wives together.

7 Conclusion and areas for future research

Gendered relations are embedded in all dimensions of food systems. In this chapter, we have reviewed some of the key contributions that enhance our understanding of how gendered relations affect the agricultural sector and how they affect the well-being of those involved in agriculture.

The growth of the literature testing household models stimulated empirical tests of these models in a range of country settings. Evidence has rejected the unitary model in favor of the collective model, with increased emphasis in recent work on testing whether households make Pareto efficient decisions. Most of this evidence comes from West Africa and needs to be confirmed in other contexts. Why does it matter whether households are efficient? It matters to development outcomes. Inefficient households don't share risk—and in situations of covariate (pandemics, climate shocks) and idiosyncratic risk, incomplete risk sharing within households could have negative consequences. The latest work on household models has identified factors that shape marriage markets as distribution factors that affect bargaining power within households. These are basic legal and cultural institutions that lay the foundation for present and future gender inequalities. Given the reach of these institutions, we need better evidence on how policies and programs affect distribution factors in a wide range of country contexts. Combined with the greater availability of individual-level data on inputs and outcomes, this is a promising area for future research.

Many of the stylized facts about women's contribution to agriculture have been challenged in recent years. Two of the most common ones are that women provide 60–80% of the labor in agriculture and that they produce 60–80% of the world's food. A challenge to the former suggests that women provide only about 40% of the labor in cropping in six African countries (Palacios-Lopez, Christiaensen, & Kilic, 2017). This suggests that women play an important role, even in this limited definition of agriculture, but do not provide the majority of the labor. An analysis of how much food is produced by women demonstrates that it would be almost impossible to disentangle food production based on the gender of who produces it. When men clear the field, women plant the seeds and weed, and they jointly harvest the crops, it is hard to assign the output to either men or women (Doss, 2014).

To further complicate issues, both plot-based and crop-based measures of women's contribution to agriculture often omit household activities related to animal agriculture. This is potentially a very important omission: by most estimates, livestock-related activities account for a large fraction of total agricultural production in almost all farming systems—and often for even larger fractions of cash income. For example, FAO (2009) reports that livestock contribute 40% of the global value of agriculture output. Livestock also play important roles as household assets, including as stores of value and as a form of insurance. It is a nontrivial omission to exclude animal agriculture from calculations of men's and women's agricultural labor. Moreover, animal care is often highly gendered—with women responsible

for dairying and small livestock in many societies, while men are typically responsible for ranching and herding. This implies that in different farming systems, the omission of animal agriculture may have important implications for the measurement of men's and women's relative agricultural productivity.

And while it is important to know how much labor women contribute, this doesn't tell us about their productivity within agriculture or about what policies we should implement to improve either agricultural productivity or women's well-being. Understanding the differential productivity on men's, women's, and jointly managed plots requires further study of the links among the decisions made by different managers within the household.

There is scope for further research on understanding how joint decision-making within households with regard to agricultural decisions influences the outcomes. Similarly, women's empowerment may change the outcomes in households where women do not farm independent plots.

Impact evaluations of value chain interventions illustrate the potential for working with market actors to test approaches for ensuring that market interventions are gender equitable. Experimental approaches may help in the design and implementation of market-oriented interventions to achieve development objectives, particularly if they help us unbundle intervention packages to see which component is most effective. Despite the increased popularity of experimental approaches in studying gender and market interactions, there is still an important role for descriptive analysis of VCs. Studies employing both quantitative and qualitative methods may be the best option for understanding gender dynamics in value chains. Moreover, studies of value chain interventions need to consider how these affect gender dynamics within the households of value chain actors.

Impact evaluations of nutrition-sensitive agricultural programs illustrate the gains from using qualitative and quantitative methods and emphasize the role of good evaluation design. This does not mean that only experimental approaches are valid. Indeed, limiting impact evaluation only to programs that are amenable to randomization may exclude programs that have the potential to affect a larger number of people but that cannot be randomized. Good design for gender-informed evaluations should have a credible counterfactual, the ability to unbundle complex interventions, and should be based on an understanding of gender norms in the project area. Better still, the intervention should be designed to take those gender norms into account, both at the pilot stage, and when scaling up. Does gender inequality in the use, control, and ownership of assets affect takeup of agricultural interventions? Do baseline inequalities influence the distribution of benefits from the intervention? Although these seem like basic questions, they are often neglected in the design and evaluation of agricultural development programs.

Innovations in data collection have recently greatly increased the data available for gender analysis in agricultural research. Yet this increased availability of data also poses some new challenges. The evidence is clear that responses to a variety of survey questions differ depending on who we ask, but additional research is

needed to understand what we can learn from these different responses. Under what circumstances does including women's voices in the data change our findings and policy implications? The early work on gender and intrahousehold issues often used data that was inadequate for the intended purpose, but demonstrated that better data was needed because the questions were important and new insights were needed. Now that better data exist, the challenge is to focus on the research areas with potentially high returns for understanding agricultural households.

Ultimately, the research questions will determine what data is needed. With innovations in data collection and the increased availability of data at both the individual and household level, it is increasingly possible to match appropriate data to a given research question, and, if needed, to design new data collection accordingly. When we conduct analyses at the individual level, regarding farmers, workers or students, for example, it is critical to ensure that the sample includes both men and women. For household level analyses, for example on the household farm enterprise, data about individuals within the household can be useful, such as women's level of empowerment or labor inputs by gender. To understand how intrahousehold relations affect agricultural production, we would need data from both men and women in the same households. Using mixed methods approaches and drawing on qualitative analyses to shape the econometric analysis can not only ensure that the quantitative approach reflects the local context but may also shape how we interpret the findings.

Most of the studies we have reviewed in this chapter have focused on the individual and the household. Gender relations can also influence outcomes on a higher scale. One of these is at the level of the collective, or groups and organizations. Given the importance of groups and social networks to many development outcomes, as well as the use of group-based approaches to service and intervention delivery, such as through self-help groups and farmer organizations, we need to understand better how gender relations work through groups and how to measure collective agency. Another process that is occurring at a higher scale is climate change. How does climate change differentially impact individuals, by gender, and how does gender affect adaptation to climate change? This is an area of work that deserves future exploration.

As this extensive body of work demonstrates, to understand how food systems are changing, it is important to consider not just the aggregate patterns but to understand who is involved and who is affected. Who makes the decisions, provides the labor, goes to market, processes the crops and eats the food? We have focused on the gender dimensions because they continue to be neglected in many analyses. Increasingly, evidence suggests that gender is not the only factor that should be taken into consideration. Gender intersects age, marital status, social class, and race/ethnicity, among others, to both shape the food systems and who benefits from them. Transforming food systems to be more equitable and inclusive requires paying attention to these intersecting dimensions in research, programs, and policy. This is a promising and exciting area for future work.

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